

Measuring connectivity

- **Edge Connectivity**

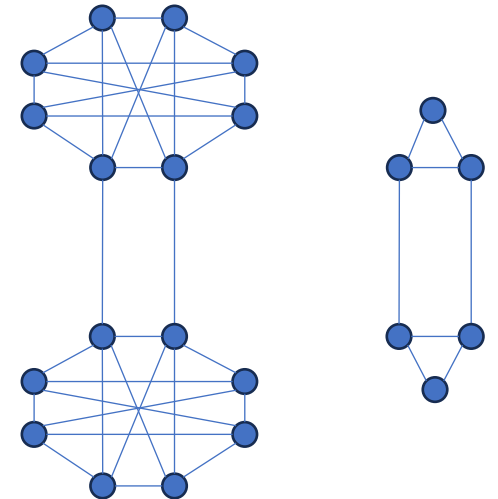
- Minimum number of edges whose removal disconnects the graph.
- Captures global robustness against edge failures.

- **Vertex Connectivity**

- Minimum number of vertices whose removal disconnects the graph.
- Captures global robustness against vertex failures.

Key limitations of Edge & Vertex Connectivity:

- Ignore the size of the affected region.
- Larger cities → more roads



Note: We focus on **simple unweighted graphs**, but the definitions also extend to weighted graphs with multi-edges and self-loops.

Conductance

Consider a graph $G = (V, E)$.

Volume of a vertex set S :

- $\text{vol}(S) = \sum_{v \in S} \deg(v)$.

Conductance of a cut $(S, V \setminus S)$:

- $\Phi(S) = \frac{|E(S, V \setminus S)|}{\min\{\text{vol}(S), \text{vol}(V \setminus S)\}}$, where $E(A, B) = \{ \{u, v\} \in E \mid u \in A \text{ and } v \in B \}$.

It measures the ratio between the size of a cut and the size of the region it separates.

Some facts:

- If $A \cap B = \emptyset$, then $\text{vol}(A \cup B) = \text{vol}(A) + \text{vol}(B)$.
- $0 \leq \Phi(S) \leq 1$.

Note: We focus on **simple unweighted graphs**, but the definitions also extend to weighted graphs with multi-edges and self-loops.

Conductance

Consider a graph $G = (V, E)$.

Some facts:

- $0 \leq \Phi(G) \leq 1$.
- If $G = (V, E)$ is connected, then $\Phi(G) \geq 1/|E|$.

Volume of a vertex set S :

- $\text{vol}(S) = \sum_{v \in S} \deg(v)$.

Conductance of a cut $(S, V \setminus S)$:

- $\Phi(S) = \frac{|E(S, V \setminus S)|}{\min\{\text{vol}(S), \text{vol}(V \setminus S)\}}$, where $E(A, B) = \{ \{u, v\} \in E \mid u \in A \text{ and } v \in B \}$.

Conductance of a graph G :

- $\Phi(G) = \min_{S \subseteq V : S \neq V \text{ and } S \neq \emptyset} \Phi(S)$.

The minimum conductance over all cuts in the graph.

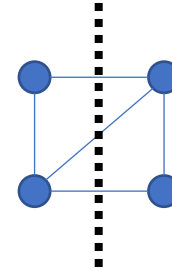
Intuition:

- What is the weakest cut in the network?
- Which region is easiest to separate relative to its size?

Conductance

Volume of a vertex set S :

- $\text{vol}(S) = \sum_{v \in S} \deg(v)$.



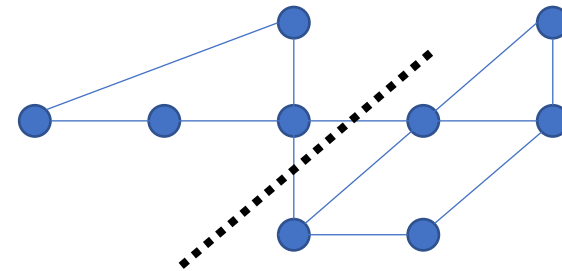
$$\text{Conductance} = \frac{3}{\min\{2+3, 2+3\}} = \frac{3}{5} = 0.6$$

Conductance of a cut $(S, V \setminus S)$:

- $\Phi(S) = \frac{|E(S, V \setminus S)|}{\min\{\text{vol}(S), \text{vol}(V \setminus S)\}}$, where $E(A, B) = \{ \{u, v\} \in E \mid u \in A \text{ and } v \in B \}$.

Conductance of a graph G :

- $\Phi(G) = \min_{S \subseteq V : S \neq V \text{ and } S \neq \emptyset} \Phi(S)$.



$$\text{Conductance} = \frac{2}{\min\{4+2+2+2, 4+3+3+2+2\}} = \frac{2}{10} = 0.2$$

Expanders

Two $(n/2)$ -cliques connected by an edge

- A graph G is a **ϕ -expander** $\leftrightarrow \Phi(G) \geq \phi$.

Graph	Conductance
Dumbbells	$\Theta(1/n^2)$
Paths	$\Theta(1/n)$
Cycles	$\Theta(1/n)$
$(\sqrt{n} \times \sqrt{n})$ -grids	$\Theta(1/\sqrt{n})$
Hypercubes	$\Theta(1/\log n)$
Stars	$\Theta(1)$
Complete graphs / cliques	$\Theta(1)$

Exercises!

Key takeaways:

- Conductance provides a size-sensitive notion of expansion.
- It better captures robustness and connectivity in real networks.

Comparing with vertex & edge connectivity

Robustness against edge deletions

- Suppose we remove a subset of edges $D \subseteq E$ from a ϕ -expander $G = (V, E)$.
 - The resulting graph is $G' = (V, E \setminus D)$.
 - We say that a connected component S of G' is **small** if $\text{vol}_G(S) \leq \frac{\text{vol}_G(V)}{2}$.
- **Claim:** The total volume of small connected components of G' is $O\left(\frac{|D|}{\phi}\right)$.

In other words, deleting the edges $D \subseteq E$ can only disconnect a small subset of volume $O\left(\frac{|D|}{\phi}\right)$.

- Therefore, expanders are **robust against edge deletions**.

Useful in fault-tolerant distributed computing

Robustness against edge deletions

- Suppose we remove a subset of edges $D \subseteq E$ from a ϕ -expander $G = (V, E)$.
 - The resulting graph is $G' = (V, E \setminus D)$.
 - We say that a connected component S of G' is **small** if $\text{vol}_G(S) \leq \frac{\text{vol}_G(V)}{2}$.
- **Claim:** The total volume of small connected components of G' is $O\left(\frac{|D|}{\phi}\right)$.

Proof of the claim – Case 1: There is a connected component that is not small.

- Let X be the union of all small connected components.

$$\begin{array}{ccccc} \boxed{\text{vol}_G(X) \leq \frac{\text{vol}_G(V)}{2}} & \longrightarrow & \boxed{\phi \leq \frac{|E(X, V \setminus X)|}{\text{vol}_G(X)} \leq \frac{|D|}{\text{vol}_G(X)}} & \longrightarrow & \boxed{\text{vol}_G(X) \leq \frac{|D|}{\phi}} \end{array}$$

Robustness against edge deletions

Proof of the claim – Case 2: All connected components are small.

- We show that there is $X = \text{union of } \underline{\text{some}} \text{ small connected components}$ such that:
 - $\frac{\text{vol}_G(V)}{3} \leq \text{vol}_G(X) \leq \frac{2\text{vol}_G(V)}{3}$

$$\phi \leq \frac{|E(X, V \setminus X)|}{\min\{\text{vol}_G(X), \text{vol}_G(V \setminus X)\}} \leq \frac{|D|}{\text{vol}_G(V)/3}$$

$$\min\{\text{vol}_G(X), \text{vol}_G(V \setminus X)\} \geq \frac{\text{vol}_G(V)}{3}$$

$$\text{The total volume of small connected components} = \text{vol}_G(V) \leq \frac{3|D|}{\phi}$$

Robustness against edge deletions

Proof of the claim – Case 2: All connected components are small.

- We show that there is $X = \text{union of } \underline{\text{some}} \text{ small connected components}$ such that:
 - $\frac{\text{vol}_G(V)}{3} \leq \text{vol}_G(X) \leq \frac{2\text{vol}_G(V)}{3}$

Selection of X :

- If there is a small connected component S with $\text{vol}_G(S) \geq \frac{\text{vol}_G(V)}{3}$:
 - We may select $X \leftarrow S$.
- Otherwise:
 - Initialize $X \leftarrow \emptyset$.
 - While $(\text{vol}_G(X) < \frac{\text{vol}_G(V)}{3})$
 - $X \leftarrow \text{union of } X \text{ and some small connected component.}$

Robustness against edge deletions

- Suppose we remove a subset of edges $D \subseteq E$ from a ϕ -expander $G = (V, E)$.
 - The resulting graph is $G' = (V, E \setminus D)$.
 - We say that a connected component S of G' is **small** if $\text{vol}_G(S) \leq \frac{\text{vol}_G(V)}{2}$.
- **Claim:** The total volume of small connected components of G' is $O\left(\frac{|D|}{\phi}\right)$.

Next: We will demonstrate an application of this result to **sublinear-time algorithms**.

Fast connectivity check under edge failures

- **Input:**

- A ϕ -expander $G = (V, E)$.
- A subset of edges $D \subseteq E$.
- Two vertices s and t .

- **Goal:**

- Decide whether s and t are connected in $G' = (V, E \setminus D)$.

- **Claim:**

- The problem can be solved in $O\left(\frac{|D|}{\phi}\right)$ time.

It suffices to do BFS to explore up to $O\left(\frac{|D|}{\phi}\right)$ volume from both s and t .

The volume is sufficient for us to decide whether s and t belong to small connected components.

If both s and t are not in small connected components, then the algorithm returns **YES**.

Otherwise, at least one of s and t belongs to a small connected component, and the volume is sufficient for us to search the entire component to decide whether s and t are connected.

Expanders have small diameter

- **Claim:** The diameter of any n -vertex ϕ -expander is $O\left(\frac{\log n}{\phi}\right)$.



Maximum shortest-path distance between any two vertices

Intuition:

- In a well-connected network:
 - Everything is only a few steps away.

Algorithmic Advantages:

- Quick broadcast and aggregation:
 - Messages reach the whole network rapidly.

Expanders have small diameter

- **Claim:** The diameter of any n -vertex ϕ -expander is $O\left(\frac{\log n}{\phi}\right)$.

We prove the claim using a **ball-growing** argument:

- $B(v, r)$ = the ball of radius r around v .
- It suffices to show that:
 - There exists $r^* \in O\left(\frac{\log n}{\phi}\right)$ such that $\text{vol}(B(v, r^*)) > \frac{\text{vol}(V)}{2}$ for every vertex v .



$B(u, r^*) \cap B(v, r^*) \neq \emptyset$ for any u and v .



Graph diameter $\leq 2r^* \in O\left(\frac{\log n}{\phi}\right)$.

Expanders have small diameter

- **Claim:** The diameter of any n -vertex ϕ -expander is $O\left(\frac{\log n}{\phi}\right)$.

We prove the claim using a **ball-growing** argument:

- $B(v, r)$ = the ball of radius r around v .

If $\text{vol}(B(v, r)) \leq \frac{\text{vol}(V)}{2}$, then $\text{vol}(B(v, r + 1)) \geq (1 + \phi)\text{vol}(B(v, r))$.

$$\begin{aligned} & \text{vol}(B(v, r + 1)) \\ & \geq \text{vol}(B(v, r)) + E(B(v, r), V \setminus B(v, r)) \\ & \geq (1 + \phi)\text{vol}(B(v, r)) \end{aligned}$$

Expanders have small diameter

- **Claim:** The diameter of any n -vertex ϕ -expander is $O\left(\frac{\log n}{\phi}\right)$.

We prove the claim using a **ball-growing** argument:

- $B(v, r)$ = the ball of radius r around v .

$$\text{If } \text{vol}(B(v, r)) \leq \frac{\text{vol}(V)}{2}, \text{ then } \text{vol}(B(v, r+1)) \geq (1 + \phi) \text{vol}(B(v, r)).$$



$$\text{If } \text{vol}(B(v, r)) \leq \frac{\text{vol}(V)}{2}, \text{ then } \text{vol}(B(v, r)) > e^{\frac{r\phi}{2}}.$$

$$\begin{aligned} \text{vol}(B(v, r)) &\geq (1 + \phi)^r \text{vol}(B(v, 0)) \geq (1 + \phi)^r \\ &> e^{\frac{r^2\phi}{r\phi+r}} \geq e^{\frac{r^2\phi}{2r}} = e^{\frac{r\phi}{2}} \end{aligned}$$

$$\left(1 + \frac{x}{y}\right)^y > e^{\frac{xy}{x+y}}$$

Expanders have small diameter

- **Claim:** The diameter of any n -vertex ϕ -expander is $O\left(\frac{\log n}{\phi}\right)$.

We prove the claim using a **ball-growing** argument:

- $B(v, r)$ = the ball of radius r around v .

If $\text{vol}(B(v, r)) \leq \frac{\text{vol}(V)}{2}$, then $\text{vol}(B(v, r+1)) \geq (1 + \phi)\text{vol}(B(v, r))$.

If $\text{vol}(B(v, r)) \leq \frac{\text{vol}(V)}{2}$, then $\text{vol}(B(v, r)) > e^{\frac{r\phi}{2}}$. $\rightarrow e^{\frac{r\phi}{2}} < \frac{\text{vol}(V)}{2} \rightarrow r < \frac{2 \ln \frac{\text{vol}(V)}{2}}{\phi}$

There exists $r^* \in O\left(\frac{\log n}{\phi}\right)$ such that $\text{vol}(B(v, r^*)) > \frac{\text{vol}(V)}{2}$.

Construction of expanders

- **Stars** and **cliques** already have very good conductance:
 - $\Phi(G) \in \Omega(1)$
- However, they are undesirable in that they have high-degree vertices.
- Can we simultaneously achieve both of the following?
 - $\Phi(G) \in \Omega(1)$
 - Maximum degree $\in O(1)$
- **Hypercubes** nearly achieve this goal, up to a factor of $O(\log n)$.

Randomized construction 1

- Erdős-Renyi random graph $\mathcal{G}(n, p)$:
 - An n -vertex graph $G = (V, E)$ such that:
 - for each pair $\{u, v\}$ of vertices in V , $\{u, v\} \in E$ with probability p independently.
- There is a choice of sampling probability $p \in \Theta\left(\frac{\log n}{n}\right)$ such that:
 - With probability $1 - 1/\text{poly}(n)$,
 - The maximum degree of G is $O(\log n)$.
 - $\Phi(G) \in \Omega(1)$.

Intuitively, this means that almost all graphs are good expanders!

This is an **exercise**:

- Apply a Chernoff bound for every cut.
- Sum up the error probability.

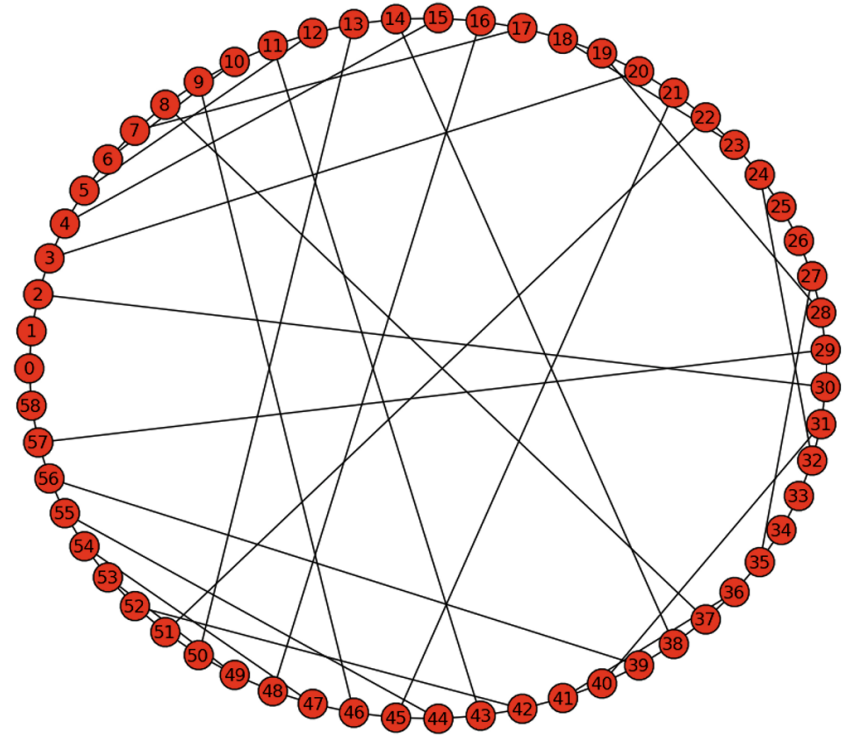
Randomized construction 2

- Generate d random perfect matching over n vertices.
- Take the union of all edges in the matchings.
- There exists a constant d such that the resulting graph G satisfies $\Phi(G) \in \Omega(1)$ with probability $1 - 1/\text{poly}(n)$.

<https://lucatrevisan.github.io/teaching/expanders2016/lecture19.pdf>

Deterministic construction

- Let p be a prime.
- Define the graph $G_p = (V_p, E_p)$ as follows.
 - $V_p = \{0, 1, \dots, p-1\}$
 - Each vertex $a \in V_p \setminus \{0\}$ is connected to:
 - $a - 1 \bmod p$
 - $a + 1 \bmod p$
 - a^{-1} (multiplicative inverse: the unique element $a^{-1} \in V_p$ such that $a \cdot a^{-1} \bmod p = 1$)
 - The vertex 0 is connected to 1, to $p-1$, and has a self-loop.
- $\Phi(G_p) \in \Omega(1)$
- The degree of every vertex in G_p equals 3.



Variations

Claim: If the maximum degree Δ of G is $O(1)$, then $\Phi(G)$, $h(G)$, $h_{\text{in}}(G)$, $h_{\text{out}}(G)$ are within a constant factor of each other.

- Given a vertex set $S \subseteq V$, define:
 - $\partial_{\text{in}}(S)$ = the set of vertices in S adjacent to $V \setminus S$.
 - $\partial_{\text{out}}(S)$ = the set of vertices in $V \setminus S$ adjacent to S .

- Edge expansion:**

- $h(G) = \min_{0 < |S| \leq \frac{n}{2}} \frac{|E(S, V \setminus S)|}{|S|}$

- Vertex expansion:**

- $h_{\text{in}}(G) = \min_{0 < |S| \leq \frac{n}{2}} \frac{|\partial_{\text{in}}(S)|}{|S|}$

- $h_{\text{out}}(G) = \min_{0 < |S| \leq \frac{n}{2}} \frac{|\partial_{\text{out}}(S)|}{|S|}$

What about $\Phi(G)$?

$h(G)$, $h_{\text{in}}(G)$, $h_{\text{out}}(G)$ are within a constant factor of each other.

- $|\partial_{\text{in}}(S)| \leq |E(V, V \setminus S)| \leq \Delta |\partial_{\text{in}}(S)|$
- $|\partial_{\text{out}}(S)| \leq |E(V, V \setminus S)| \leq \Delta |\partial_{\text{out}}(S)|$

Variations

Claim: If the maximum degree Δ of G is $O(1)$, then $\Phi(G)$, $h(G)$, $h_{\text{in}}(G)$, $h_{\text{out}}(G)$ are within a constant factor of each other.

- It remains to show that the two parameters are within an $O(1)$ -factor:
 - $h(G) = \min_{0 < |S| \leq \frac{n}{2}} \frac{|E(S, V \setminus S)|}{|S|}$
 - $\Phi(G) = \min_{S \subseteq V : S \neq V \text{ and } S \neq \emptyset} \frac{|E(S, V \setminus S)|}{\min\{\text{vol}(S), \text{vol}(V \setminus S)\}} = \min_{0 < |S| \leq \frac{n}{2}} \frac{|E(S, V \setminus S)|}{\min\{\text{vol}(S), \text{vol}(V \setminus S)\}}$
- To do so, it suffices to show that for any $0 < |S| \leq \frac{n}{2}$, the two parameters are within an $O(1)$ -factor:
 - $|S|$
 - $\min\{\text{vol}(S), \text{vol}(V \setminus S)\}$
- Indeed, $|S| \leq \min\{|S|, |V \setminus S|\} \leq \min\{\text{vol}(S), \text{vol}(V \setminus S)\} \leq \text{vol}(S) \leq \Delta|S|$.
 - This step requires that there is no isolated vertex.
 - If there is an isolated vertex, then all the parameters $\Phi(G)$, $h(G)$, $h_{\text{in}}(G)$, $h_{\text{out}}(G)$ are zero.

Vertex expansion vs. conductance

- We have shown the following results for ϕ -expanders:
 - Robustness against edge deletions.
 - Small diameter.
- **Exercise:** Extend the above results to graphs with high vertex expansion:
 - Robustness against vertex deletions.
 - Small diameter.

Vertex expansion vs. conductance

- Vertex expansion and conductance can be very different!

There is an n -vertex graph G with

- $\Phi(G) \in \Omega(1)$
- $h_{\text{in}}(G) \in O\left(\frac{1}{n}\right)$
- $h_{\text{out}}(G) \in O\left(\frac{1}{n}\right)$

There is an n -vertex graph G with

- $\Phi(G) \in O\left(\frac{1}{n}\right)$
- $h_{\text{in}}(G) \in \Omega(1)$
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Vertex expansion vs. conductance

- Vertex expansion and conductance can be very different!

There is an n -vertex graph G with

- $\Phi(G) \in \Omega(1)$
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- $h_{\text{out}}(G) \in O\left(\frac{1}{n}\right)$

Star graphs

There is an n -vertex graph G with

- $\Phi(G) \in O\left(\frac{1}{n}\right)$
- $h_{\text{in}}(G) \in \Omega(1)$
- $h_{\text{out}}(G) \in \Omega(1)$

Two $(n/2)$ -cliques with a perfect matching between them

Computing conductance

- **Recall:**

- Conductance of a graph measures how well-connected it is.
- Graphs with high conductance are robust against failures.

- It is **NP-hard** to compute the conductance exactly.

- **Next:** $O(\log n)$ -approximation in polynomial time. The Leighton–Rao algorithm



Computing an estimate ϕ such that $\Phi(G) \leq \phi \leq \alpha \cdot \Phi(G)$ for some $\alpha \in O(\log n)$.

Tool 1: LP relaxation

- A **general strategy** for **approximation algorithms** for an NP-hard problem:

- Relax it to a problem that can be solved in polynomial time.

↑
In our case, linear programming (LP).

- Apply an efficient algorithm to solve the relaxed problem.

- Transform the solution back to a valid solution of the original problem.

↑
This step incurs some approximation error.

- Maximizing/minimizing a linear objective function.
 - Subject to linear constraints (equalities and inequalities).
- Variables take real values.

Metric space

- A **metric space** is an ordered pair (X, d) , where
 - X is a set.
 - $d: X \times X \rightarrow \mathbb{R}_{\geq 0}$ is a **distance function (metric)** on X meeting the conditions:
 - $d(x, x) = 0$
 - $d(x, y) = d(y, x)$ (symmetry)
 - $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality)

Note: In mathematics, the definition of a metric space includes an additional requirement:

- if $x \neq y$, then $d(x, y) \neq 0$.

Without it, it is known as a pseudometric or semimetric space.

In computer science, it is more common to simply drop this requirement from the definition of metric spaces.

Metric space

- A **metric space** is an ordered pair (X, d) , where
 - X is a set.
 - $d: X \times X \rightarrow \mathbb{R}_{\geq 0}$ is a **distance function (metric)** on X meeting the conditions:
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 - $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality)

Closure under addition:

- If d_1 and d_2 are metrics on the same set X , then:
 - $d_1 + d_2$ is also a metric on X .

Closure under scalar multiplication:

- If d is a metric on X and $\alpha \geq 0$ is a scalar, then:
 - $\alpha \cdot d$ is also a metric on X .

Line metric

- **Line metric:**

- Given a function $f: X \rightarrow \mathbb{R}$, define

$$d(x, y) = |f(x) - f(y)|$$

ℓ_1 metric

- **Line metric:**

- Given a function $f: X \rightarrow \mathbb{R}$, define
$$d(x, y) = |f(x) - f(y)|$$

- **ℓ_1 metric:**

- Given a function $f: X \rightarrow \mathbb{R}^k$, define

$$d(x, y) = \|f(x) - f(y)\|_1 = \sum_{i=1}^k |f_i(x) - f_i(y)|$$

Observation: A k -dimensional ℓ_1 metric is the sum of k line metrics.

Cut metric

Observation: A line metric is a linear combination of $|X| - 1$ cut metrics.

- **Line metric:**

- Given a function $f: X \rightarrow \mathbb{R}$, define

$$d(x, y) = |f(x) - f(y)|$$

Order $X = \{x_1, \dots, x_n\}$ so that $f(x_1) \leq \dots \leq f(x_n)$.

- Set $\alpha_i = f(x_{i+1}) - f(x_i)$
- Set d_i to be a cut metric with $S = \{x_1, \dots, x_i\}$.
- Then $d = \sum_{i=1}^{n-1} \alpha_i d_i$.

- **Cut metric:**

- Given a subset $S \subseteq X$, define

$$d(x, y) = \begin{cases} 1 & \text{if } ((x \in S) \wedge (y \in X \setminus S)) \vee ((x \in X \setminus S) \wedge (y \in S)) \\ 0 & \text{otherwise} \end{cases}$$

Observation: Cut metric is a special case of line metric, by setting $f(x) = \begin{cases} 1 & \text{if } x \in S \\ 0 & \text{otherwise} \end{cases}$

Cut metric

Observation: A line metric is a linear combination of $|X| - 1$ cut metrics.

+

Observation: A k -dimensional ℓ_1 metric is the sum of k line metrics.



Observation: A k -dimensional ℓ_1 metric is a linear combination of $k(|X| - 1)$ cut metrics.

Shortest path metric

- **Shortest path metric:**

- Given a graph $G = (X, E)$, define

$d(x, y)$ = shortest path distance between x and y

Tool 2: metric embedding

- **Motivation:** Computing the distance in some metrics is more expensive than in other metrics.

Embed a complicated metric into a simpler one with **small distortion**.

Greatly reduce the cost of distance computation, at the expense of only a **small approximation error**.

Next: Bourgain's theorem allows us to embed any n -point metric into an $O(\log^2 n)$ -dimensional ℓ_1 metric space with $O(\log n)$ distortion.

A linear combination of $O(n \log^2 n)$ **cut metrics**!

Tool 2: metric embedding

- **Bourgain's theorem for metric embeddings:**

- Given any n -point metric space (X, d) ,
 - there is an embedding $f: X \rightarrow \mathbb{R}^{O(\log^2 n)}$ such that for any $x, y \in X$,
 - $\|f(x) - f(y)\|_1 \leq d(x, y) \leq O(\log n) \cdot \|f(x) - f(y)\|_1$



If $d(x, y)$ is given for all $x, y \in X$, then the mapping f can be computed in (randomized) **polynomial time**.

A quick application

- **Bourgain's theorem for metric embeddings:**

- Given any n -point metric space (X, d) ,
 - there is an embedding $f: X \rightarrow \mathbb{R}^{O(\log^2 n)}$ such that for any $x, y \in X$,
 - $\|f(x) - f(y)\|_1 \leq d(x, y) \leq O(\log n) \cdot \|f(x) - f(y)\|_1$



Apply Bourgain's theorem to the shortest path metric.



We can label each vertex with a vector of $O(\log^2 n)$ values so that:

- Given any two vertices u and v :
 - We can obtain an $O(\log n)$ -approximation of $\text{dist}(u, v)$ in **only** $O(\log^2 n)$ time by reading the labels of u and v .

Generalized conductance

with positive edge weights

- Let $G = (V, E_G)$ and $H = (V, E_H)$ be **weighted** graphs on the same vertex set V .

$$\Phi(G) = \min_{S: S \neq V \text{ and } S \neq \emptyset} \frac{|E_G(S, V \setminus S)|}{\min\{\text{vol}_G(S), \text{vol}_G(V \setminus S)\}} \quad \Phi(G, H) = \min_{S: E_H(S, V \setminus S) \neq \emptyset} \frac{|E_G(S, V \setminus S)|}{|E_H(S, V \setminus S)|}$$

Here we slightly abuse the notation to write:

- $|E_G(S, V \setminus S)| = \sum_{e \in E_G(S, V \setminus S)} w_G(e)$
- $|E_H(S, V \setminus S)| = \sum_{e \in E_H(S, V \setminus S)} w_H(e)$
- $\text{vol}_G(S) = \sum_{v \in S} \sum_{e: v \in e} w_G(e)$

Generalized conductance

- Let $G = (V, E_G)$ and $H = (V, E_H)$ be weighted graphs on the same vertex set V .

$$\Phi(G) = \min_{S: S \neq V \text{ and } S \neq \emptyset} \frac{|E_G(S, V \setminus S)|}{\min\{\text{vol}_G(S), \text{vol}_G(V \setminus S)\}} \quad \Phi(G, H) = \min_{S: E_H(S, V \setminus S) \neq \emptyset} \frac{|E_G(S, V \setminus S)|}{|E_H(S, V \setminus S)|}$$

Claim: Given G , there is a choice of H such that $\Phi(G)$ and $\Phi(G, H)$ are within a constant factor.

$$\frac{|E_G(S, V \setminus S)|}{\min\{\text{vol}_G(S), \text{vol}_G(V \setminus S)\}} \xleftrightarrow{\text{within a constant factor}} \frac{|E_G(S, V \setminus S)|}{\left(\frac{\text{vol}_G(S) \cdot \text{vol}_G(V \setminus S)}{\text{vol}_G(V)}\right)}$$

$$\frac{\text{vol}_G(S) \cdot \text{vol}_G(V \setminus S)}{\text{vol}_G(V)} = \min\{\text{vol}_G(S), \text{vol}_G(V \setminus S)\} \cdot \overbrace{\frac{\max\{\text{vol}_G(S), \text{vol}_G(V \setminus S)\}}{\text{vol}_G(V)}}^{\in [0.5, 1]}$$

Generalized conductance

- Let $G = (V, E_G)$ and $H = (V, E_H)$ be weighted graphs on the same vertex set V .

$$\Phi(G) = \min_{S: S \neq V \text{ and } S \neq \emptyset} \frac{|E_G(S, V \setminus S)|}{\min\{\text{vol}_G(S), \text{vol}_G(V \setminus S)\}} \quad \Phi(G, H) = \min_{S: E_H(S, V \setminus S) \neq \emptyset} \frac{|E_G(S, V \setminus S)|}{|E_H(S, V \setminus S)|}$$

Claim: Given G , there is a choice of H such that $\Phi(G)$ and $\Phi(G, H)$ are within a constant factor.

$$\frac{|E_G(S, V \setminus S)|}{\min\{\text{vol}_G(S), \text{vol}_G(V \setminus S)\}} \xleftrightarrow{\text{within a constant factor}} \frac{|E_G(S, V \setminus S)|}{\left(\frac{\text{vol}_G(S) \cdot \text{vol}_G(V \setminus S)}{\text{vol}_G(V)}\right)} \xleftrightarrow{w_H(u, v) = \frac{\deg_G(u) \cdot \deg_G(v)}{\text{vol}_G(V)}} \frac{|E_G(S, V \setminus S)|}{|E_H(S, V \setminus S)|}$$

Connection to cut metric

- Let $G = (V, E_G)$ and $H = (V, E_H)$ be weighted graphs on the same vertex set V .
- Let d_S be a cut metric for the cut S of V .

$\Phi(G, H)$

Minimize:

- $\sum_{u,v} w_G(u, v) \cdot d(u, v)$

Subject to:

- $\sum_{u,v} w_H(u, v) \cdot d(u, v) = 1$
- d is a cut metric scaled by a real value.

$$\Phi(G, H) = \min_{S : E_H(S, V \setminus S) \neq \emptyset} \frac{|E_G(S, V \setminus S)|}{|E_H(S, V \setminus S)|}$$

equivalent

$$\Phi(G, H) = \min_{S : \sum_{u,v} w_H(u, v) \cdot d_S(u, v) \neq 0} \frac{\sum_{u,v} w_G(u, v) \cdot d_S(u, v)}{\sum_{u,v} w_H(u, v) \cdot d_S(u, v)}$$

equivalent

LP relaxation

This is a relaxation in the sense that:

Any cut metric scaled by a real value is a metric.

$\text{LR}(G, H)$ considers a wider range of search space than $\Phi(G, H)$.

$$\text{LR}(G, H) \leq \Phi(G, H)$$

$\Phi(G, H)$

Minimize:

- $\sum_{u,v} w_G(u, v) \cdot d(u, v)$

Subject to:

- $\sum_{u,v} w_H(u, v) \cdot d(u, v) = 1$
- d is a cut metric scaled by a real value.

LP relaxation

$\text{LR}(G, H)$

Minimize:

- $\sum_{u,v} w_G(u, v) \cdot d(u, v)$

Subject to:

- $\sum_{u,v} w_H(u, v) \cdot d(u, v) = 1$
- d is a **metric**, i.e.,
 - $d(x, x) = 0 \ \forall x$
 - $d(x, y) = d(y, x) \ \forall x, y$
 - $d(x, z) \leq d(x, y) + d(y, z) \ \forall x, y, z$

Approximation ratio

An $O(\log n)$ -approximation of conductance can be computed in polynomial time by computing $\text{LR}(G, H)$ using a linear programming solver.

Claim: $\text{LR}(G, H) \in \Omega\left(\frac{\Phi(G, H)}{\log n}\right)$

$$\text{LR}(G, H) \leq \Phi(G, H)$$

$\Phi(G, H)$

Minimize:

- $\sum_{u,v} w_G(u, v) \cdot d(u, v)$

Subject to:

- $\sum_{u,v} w_H(u, v) \cdot d(u, v) = 1$
- d is a cut metric scaled by a real value.

LP relaxation →

$\text{LR}(G, H)$

Minimize:

- $\sum_{u,v} w_G(u, v) \cdot d(u, v)$

Subject to:

- $\sum_{u,v} w_H(u, v) \cdot d(u, v) = 1$
- d is a **metric**, i.e.,
 - $d(x, x) = 0 \ \forall x$
 - $d(x, y) = d(y, x) \ \forall x, y$
 - $d(x, z) \leq d(x, y) + d(y, z) \ \forall x, y, z$

Proof of the claim

Claim: $\text{LR}(G, H) \in \Omega\left(\frac{\Phi(G, H)}{\log n}\right)$

$$\text{LR}(G, H) = \frac{\sum_{u,v} w_G(u, v) \cdot d_{\text{OPT}}(u, v)}{\sum_{u,v} w_H(u, v) \cdot d_{\text{OPT}}(u, v)}$$

d_{OPT} = an optimal metric of

$\text{LR}(G, H)$

Minimize:

- $\sum_{u,v} w_G(u, v) \cdot d(u, v)$

Subject to:

- $\sum_{u,v} w_H(u, v) \cdot d(u, v) = 1$

- d is a **metric**, i.e.,

- $d(x, x) = 0 \ \forall x$

- $d(x, y) = d(y, x) \ \forall x, y$

- $d(x, z) \leq d(x, y) + d(y, z) \ \forall x, y, z$

Proof of the claim

Recall: d^* is a linear combination of $O(n \log^2 n)$ cut metrics.

Apply **Bourgain's theorem** to d_{OPT} .

There is an $O(\log^2 n)$ -dimensional ℓ_1 metric d^* such that:
 $d^*(u, v) \leq d_{\text{OPT}}(u, v) \leq O(\log n) \cdot d^*(u, v)$

There is a collection $\mathcal{C}$ of $O(n \log^2 n)$ cuts and coefficients $\{\alpha_S \mid S \in \mathcal{C}\}$ such that:

$$d^* = \sum_{S \in \mathcal{C}} \alpha_S d_S,$$

where d_S is the cut metric of the cut S .

$$\sum_{S \in \mathcal{C}} \alpha_S d_S(u, v) \leq d_{\text{OPT}}(u, v) \leq O(\log n) \cdot \sum_{S \in \mathcal{C}} \alpha_S d_S(u, v)$$

$$\text{LR}(G, H) = \frac{\sum_{u,v} w_G(u, v) \cdot d_{\text{OPT}}(u, v)}{\sum_{u,v} w_H(u, v) \cdot d_{\text{OPT}}(u, v)} \in \Omega\left(\frac{1}{\log n}\right) \cdot \frac{\sum_{u,v} w_G(u, v) \cdot \sum_{S \in \mathcal{C}} \alpha_S d_S(u, v)}{\sum_{u,v} w_H(u, v) \cdot \sum_{S \in \mathcal{C}} \alpha_S d_S(u, v)}$$

Proof of the claim

Claim: $\text{LR}(G, H) \in \Omega\left(\frac{\Phi(G, H)}{\log n}\right)$

Proof idea:

- Among all $O(n \log^2 n)$ cuts $S \in \mathcal{C}$:
 - Choose the sparsest one.
 - $\frac{\sum_{u,v} w_G(u,v) \cdot \sum_{S \in \mathcal{C}} \alpha_S d_S(u,v)}{\sum_{u,v} w_H(u,v) \cdot \sum_{S \in \mathcal{C}} \alpha_S d_S(u,v)} \geq \text{sparsity of the cut} \geq \Phi(G, H)$

$$\sum_{S \in \mathcal{C}} \alpha_S d_S(u, v) \leq d_{\text{OPT}}(u, v) \in O(\log n) \cdot \sum_{S \in \mathcal{C}} \alpha_S d_S(u, v)$$

$$\text{LR}(G, H) = \frac{\sum_{u,v} w_G(u, v) \cdot d_{\text{OPT}}(u, v)}{\sum_{u,v} w_H(u, v) \cdot d_{\text{OPT}}(u, v)} \in \Omega\left(\frac{1}{\log n}\right) \cdot \frac{\sum_{u,v} w_G(u, v) \cdot \sum_{S \in \mathcal{C}} \alpha_S d_S(u, v)}{\sum_{u,v} w_H(u, v) \cdot \sum_{S \in \mathcal{C}} \alpha_S d_S(u, v)} \in \Omega\left(\frac{\Phi(G, H)}{\log n}\right)$$

Proof of the claim

Claim: $\text{LR}(G, H) \in \Omega\left(\frac{\Phi(G, H)}{\log n}\right)$

$$\frac{\sum_{u,v} w_G(u, v) \cdot \sum_{S \in \mathcal{C}} \alpha_S d_S(u, v)}{\sum_{u,v} w_H(u, v) \cdot \sum_{S \in \mathcal{C}} \alpha_S d_S(u, v)} = \frac{\sum_{S \in \mathcal{C}} \alpha_S \cdot \sum_{u,v} w_G(u, v) d_S(u, v)}{\sum_{S \in \mathcal{C}} \alpha_S \cdot \sum_{u,v} w_H(u, v) d_S(u, v)} \geq \min_{S \in \mathcal{C}} \frac{\sum_{u,v} w_G(u, v) \cdot d_S(u, v)}{\sum_{u,v} w_H(u, v) \cdot d_S(u, v)} \geq \Phi(G, H)$$

$$\frac{\sum_i a_i}{\sum_i b_i} \geq \min_i \frac{a_i}{b_i}$$

$$\text{LR}(G, H) = \frac{\sum_{u,v} w_G(u, v) \cdot d_{\text{OPT}}(u, v)}{\sum_{u,v} w_H(u, v) \cdot d_{\text{OPT}}(u, v)} \in \Omega\left(\frac{1}{\log n}\right) \cdot \frac{\sum_{u,v} w_G(u, v) \cdot \sum_{S \in \mathcal{C}} \alpha_S d_S(u, v)}{\sum_{u,v} w_H(u, v) \cdot \sum_{S \in \mathcal{C}} \alpha_S d_S(u, v)} \in \Omega\left(\frac{\Phi(G, H)}{\log n}\right)$$

Correctness

Bourgain's theorem:

- Given any n -point metric space (X, d) ,
 - there is an embedding $f: X \rightarrow \mathbb{R}^{O(\log^2 n)}$ such that for any $x, y \in X$,
 $\|f(x) - f(y)\|_1 \leq d(x, y) \cdot O(\log n)$

Algorithm:

- Randomly sample $O(\log^2 n)$ subsets:
 $A_{i,j} \subseteq X \quad | \quad 1 \leq i \leq c \log n, \quad 1 \leq j \leq \log n$
- Each point in X joins $A_{i,j}$ with probability 2^{-j} independently.
- Set $f_{i,j}(x) = d(x, A_{i,j}) = \min_{y \in A_{i,j}} d(x, y)$.

$$\|f(x) - f(y)\|_1 = \sum_{i,j} |f_{i,j}(x) - f_{i,j}(y)|$$

Correctness:

With high probability, we have:

- Part 1:** $\sum_{i,j} |f_{i,j}(x) - f_{i,j}(y)| \leq d(x, y) \cdot c \log^2 n$
- Part 2:** $\sum_{i,j} |f_{i,j}(x) - f_{i,j}(y)| \in d(x, y) \cdot \Omega(\log n)$

With a scaling $f_{i,j}(x) \leftarrow f_{i,j}(x) \cdot \frac{1}{c \log^2 n}$,
this proves Bourgain's theorem.

Correctness: Part 1

Without loss of generality, assume $d(x, A_{i,j}) \geq d(y, A_{i,j})$

Triangle inequality

$$\begin{aligned} |f_{i,j}(x) - f_{i,j}(y)| &= |d(x, A_{i,j}) - d(y, A_{i,j})| \\ &= d(x, A_{i,j}) - d(y, A_{i,j}) \\ &\leq (d(x, y) + d(y, A_{i,j})) - d(y, A_{i,j}) \\ &= d(x, y) \end{aligned}$$

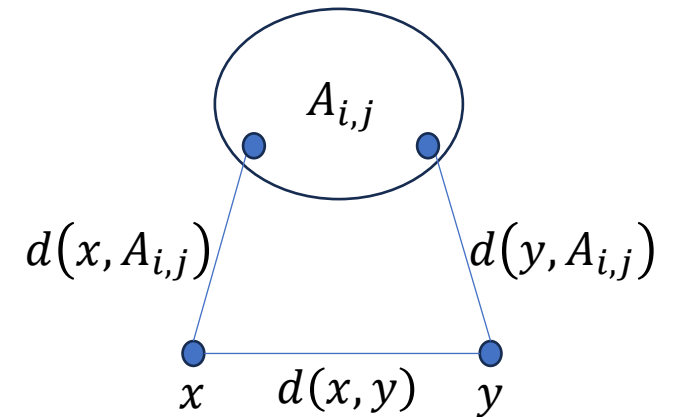
Algorithm:

- Randomly sample $O(\log^2 n)$ subsets:
 $A_{i,j} \subseteq X \quad | \quad 1 \leq i \leq c \log n, \quad 1 \leq j \leq \log n$
- Each point in X joins $A_{i,j}$ with probability 2^{-j} independently.
- Set $f_{i,j}(x) = d(x, A_{i,j}) = \min_{y \in A_{i,j}} d(x, y)$.

Correctness:

With high probability, we have:

- Part 1:** $\sum_{i,j} |f_{i,j}(x) - f_{i,j}(y)| \leq d(x, y) \cdot c \log^2 n$ ←
- Part 2:** $\sum_{i,j} |f_{i,j}(x) - f_{i,j}(y)| \in d(x, y) \cdot \Omega(\log n)$



Correctness: Part 2

Algorithm:

- Randomly sample $O(\log^2 n)$ subsets:
$$A_{i,j} \subseteq X \quad | \quad 1 \leq i \leq c \log n, \quad 1 \leq j \leq \log n$$
- Each point in X joins $A_{i,j}$ with probability 2^{-j} independently.
- Set $f_{i,j}(x) = d(x, A_{i,j}) = \min_{y \in A_{i,j}} d(x, y)$.

Correctness:

With high probability, we have:

- **Part 1:** $\sum_{i,j} |f_{i,j}(x) - f_{i,j}(y)| \leq d(x, y) \cdot c \log^2 n$
- **Part 2:** $\sum_{i,j} |f_{i,j}(x) - f_{i,j}(y)| \in d(x, y) \cdot \Omega(\log n)$ ←

Next

Main tool: Chernoff bound

- Consider the following setting.
 - $X_1, X_2, \dots, X_n$ are independent random variables taking values in $\{0, 1\}$.
 - $X = \sum_{i=1}^n X_i$.
 - $\mu = \mathbb{E}[X]$.

Chernoff bounds:

- $\Pr[X \geq (1 + \delta)\mu] \leq e^{-\frac{\delta^2 \mu}{3}}, \quad \text{for } 1 \geq \delta \geq 0.$
- $\Pr[X \geq (1 + \delta)\mu] \leq e^{-\frac{\delta^2 \mu}{2 + \delta}}, \quad \text{for } \delta \geq 0.$
- $\Pr[X \leq (1 - \delta)\mu] \leq e^{-\frac{\delta^2 \mu}{2}}, \quad \text{for } 1 \geq \delta \geq 0.$

https://en.wikipedia.org/wiki/Chernoff_bound

Key takeaway:

- X is within a constant factor of $\mathbb{E}[X]$ with a probability of $1 - e^{-\Omega(\mathbb{E}[X])}$.
- In particular, $\mathbb{E}[X] \in \Omega(c \log n)$ guarantees a success probability of $1 - n^{-\Omega(c)}$.

Correctness: Part 2

- $B(v, r)$ = the ball of radius r around v .
- $B(v, < r)$ = the open ball of radius r around v .

Observation 1:

$$|B(x, < r_j)| < 2^j \text{ or } |B(y, < r_j)| < 2^j$$

Without loss of generality, we assume this.

Observation 2:

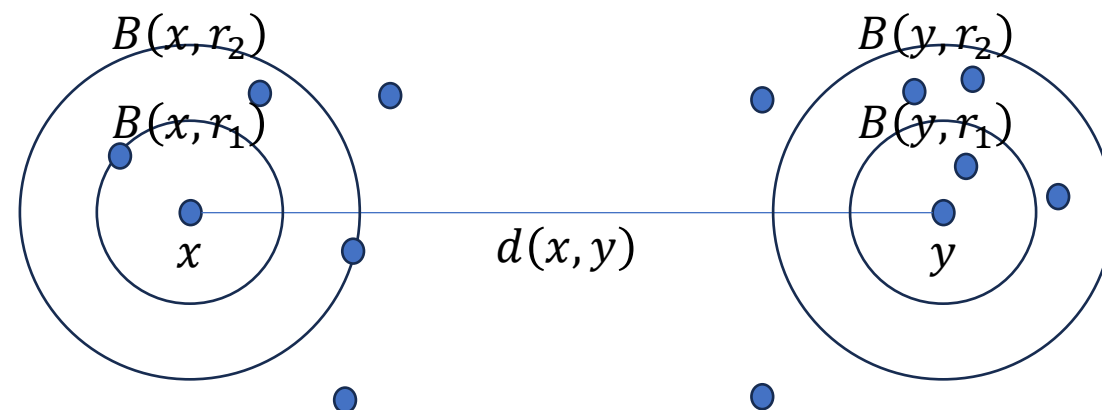
If $r_{j-1} < \frac{d(x, y)}{3}$, then

$$|B(x, r_{j-1})| \geq 2^{j-1} \text{ and } |B(y, r_{j-1})| \geq 2^{j-1}$$

Consider two points x and y in X .

r_j = the minimum of:

- Smallest number r such that $\min\{|B(x, r)|, |B(y, r)|\} \geq 2^j$
- $\frac{d(x, y)}{3}$



Correctness: Part 2

- $B(v, r)$ = the ball of radius r around v .
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Observation 1:

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$$|B(x, r_{j-1})| \geq 2^{j-1} \text{ and } |B(y, r_{j-1})| \geq 2^{j-1}$$

- Each point in X joins $A_{i,j}$ with probability 2^{-j} independently.
- Set $f_{i,j}(x) = d(x, A_{i,j}) = \min_{y \in A_{i,j}} d(x, y)$.

For $n \geq 2$, we have: $\left(1 - \frac{1}{n}\right)^n \geq \left(1 - \frac{1}{2}\right)^2 = \frac{1}{4}$

Suppose $r_{j-1} < \frac{d(x,y)}{3}$.

Let $\mathcal{E}_{i,j}$ be the event that:

- $B(x, < r_j) \cap A_{i,j} = \emptyset$
 - Probability = $\left(1 - \frac{1}{2^j}\right)^{|B(x, < r_j)|} > \left(1 - \frac{1}{2^j}\right)^{2^j} \geq \frac{1}{4}$
- $B(y, r_{j-1}) \cap A_{i,j} \neq \emptyset$
 - Probability = $1 - \left(1 - \frac{1}{2^j}\right)^{|B(y, r_{j-1})|}$

$$\geq 1 - \left(1 - \frac{1}{2^j}\right)^{2^{j-1}} \geq 1 - e^{-1/2} \geq \frac{1}{3}$$

Claim 1: $\Pr[\mathcal{E}_{i,j}] > \frac{1}{12}$

$$1 + x \leq e^x$$

Correctness: Part 2

- $B(v, r)$ = the ball of radius r around v .
- $B(v, < r)$ = the open ball of radius r around v .

Observation 1:

$$|B(x, < r_j)| < 2^j \text{ or } |B(y, < r_j)| < 2^j$$

Without loss of generality, we assume this.

Observation 2:

If $r_{j-1} < \frac{d(x,y)}{3}$, then

$$|B(x, r_{j-1})| \geq 2^{j-1} \text{ and } |B(y, r_{j-1})| \geq 2^{j-1}$$

- Each point in X joins $A_{i,j}$ with probability 2^{-j} independently.
- Set $f_{i,j}(x) = d(x, A_{i,j}) = \min_{y \in A_{i,j}} d(x, y)$.

Suppose $r_{j-1} < \frac{d(x,y)}{3}$.

Let $\mathcal{E}_{i,j}$ be the event that:

- $B(x, < r_j) \cap A_{i,j} = \emptyset$
- $B(y, r_{j-1}) \cap A_{i,j} \neq \emptyset$

Claim 1: $\Pr[\mathcal{E}_{i,j}] > \frac{1}{12}$

Claim 2: If $\mathcal{E}_{i,j}$ occurs, then:

- $|f_{i,j}(x) - f_{i,j}(y)| \geq r_j - r_{j-1}$

Correctness: Part 2

- Consider the following setting.
 - $X_i \in \{0,1\}$ is the indicator random variable for the event $\mathcal{E}_{i,j}$.
 - $X = \sum_{i=1}^{c \log n} X_i$.
 - $\mu = \mathbb{E}[X] > \frac{c \log n}{12}$.

$$\Pr\left[X \leq \frac{c \log n}{24}\right] \leq \Pr\left[X \leq \frac{\mu}{2}\right] \leq e^{-\frac{\left(\frac{1}{2}\right)^2 \mu}{2}} \in n^{-\Omega(c)}$$

With probability $1 - n^{-\Omega(c)}$, we have:

$$\sum_i |f_{i,j}(x) - f_{i,j}(y)| \geq (r_j - r_{j-1}) \cdot \frac{c \log n}{24}$$

Chernoff bound:

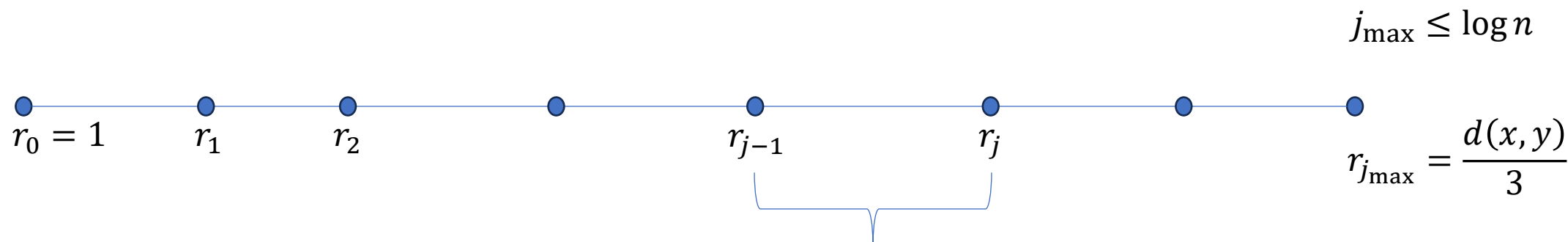
- $\Pr[X \leq (1 - \delta)\mu] \leq e^{-\frac{\delta^2 \mu}{2}}, \quad \text{for } 1 \geq \delta \geq 0.$

Claim 1: $\Pr[\mathcal{E}_{i,j}] > \frac{1}{12}$

Claim 2: If $\mathcal{E}_{i,j}$ occurs, then:

- $|f_{i,j}(x) - f_{i,j}(y)| \geq r_j - r_{j-1}$

Correctness: Part 2



With probability $1 - n^{-\Omega(c)}$, we have: $\sum_i |f_{i,j}(x) - f_{i,j}(y)| \geq (r_j - r_{j-1}) \cdot \frac{c \log n}{24}$

With probability $1 - j_{\max} \cdot n^{-\Omega(c)}$, we have: $\sum_{i,j} |f_{i,j}(x) - f_{i,j}(y)| \geq \frac{d(x,y)}{3} \cdot \frac{c \log n}{24}$



With probability $1 - \binom{n}{2} \cdot j_{\max} \cdot n^{-\Omega(c)}$, we have: $\sum_{i,j} |f_{i,j}(x) - f_{i,j}(y)| \geq \frac{d(x,y)}{3} \cdot \frac{c \log n}{24}$ for all x and y .

Can make this $1 - \frac{1}{\text{poly}(n)}$ with an arbitrarily large exponent by selecting c to be a sufficiently large constant.

This finishes the proof of correctness.

Summary

Main tool: Chernoff bound.

- $O(\log^2 n)$ subsets are needed because:
 - Need to try all $O(\log n)$ sampling probabilities: $1, \frac{1}{2}, \frac{1}{4}, \dots, \frac{1}{n}$.
 - For each of them, need to repeat for $O(\log n)$ times to get a “with high probability” guarantee.

If we are happy with an “in expectation” guarantee, then this is not needed!

$$\mathbb{E}[\|f(x) - f(y)\|_1] \leq d(x, y) \in O(\log n) \cdot \mathbb{E}[\|f(x) - f(y)\|_1]$$

Matrix representation of graphs

- Let $G = (V, E)$ be a graph, with $V = \{1, 2, \dots, n\}$.

The adjacency matrix A is an $(n \times n)$ -matrix:

$$A[i, j] = \begin{cases} 0, & \{i, j\} \notin E \\ 1, & \{i, j\} \in E \end{cases}$$

The degree matrix D is an $(n \times n)$ -matrix:

$$D[i, j] = \begin{cases} \deg(i), & i = j \\ 0, & i \neq j \end{cases}$$

The Laplacian matrix: $L = D - A$

The normalized Laplacian matrix: $N = D^{-1/2} L D^{-1/2}$

Connection to cuts

- Let $G = (V, E)$ be a graph, with $V = \{1, 2, \dots, n\}$.

The adjacency matrix A is an $(n \times n)$ -matrix:

$$A[i, j] = \begin{cases} 0, & \{i, j\} \notin E \\ 1, & \{i, j\} \in E \end{cases}$$

The degree matrix D is an $(n \times n)$ -matrix:

$$D[i, j] = \begin{cases} \deg(i), & i = j \\ 0, & i \neq j \end{cases}$$

The Laplacian matrix: $L = D - A$

The normalized Laplacian matrix: $N = D^{-1/2} L D^{-1/2}$

Let $\mathbf{x} \in \mathbb{R}^n$ be a column vector.

$$\mathbf{x}^\top L \mathbf{x} = \sum_{\{u, v\} \in E} (x_u - x_v)^2$$

If $\mathbf{x} = \mathbf{1}_S$ is the indicator vector for a cut $S \subseteq V$, then $\mathbf{x}^\top L \mathbf{x} = |E(S, V \setminus S)|$ is the size of the cut.

Regular graphs

- Let $G = (V, E)$ be a d -regular graph, with $V = \{1, 2, \dots, n\}$.

The adjacency matrix A is an $(n \times n)$ -matrix:

$$A[i, j] = \begin{cases} 0, & \{i, j\} \notin E \\ 1, & \{i, j\} \in E \end{cases}$$

The degree matrix D is an $(n \times n)$ -matrix:

$$D[i, j] = dI = \begin{cases} d, & i = j \\ 0, & i \neq j \end{cases}$$

The Laplacian matrix:

$$L = D - A = dI - A$$

The normalized Laplacian matrix:

$$N = D^{-1/2} L D^{-1/2} = I - \frac{1}{d} A$$

For simplicity, we restrict our attention to regular graphs, noting that all results extend to general graphs.

$$\mathbf{x}^\top L \mathbf{x} = \sum_{\{u, v\} \in E} (x_u - x_v)^2$$

$$\mathbf{x}^\top N \mathbf{x} = \frac{1}{d} \sum_{\{u, v\} \in E} (x_u - x_v)^2$$

Symmetric real matrices

- Let $\mathbf{M}$ be a real symmetric $(n \times n)$ -matrix.

Theorem: There exist

- $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{R}$ (they are called eigenvalues)
 - Orthonormal vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbb{R}^n$ (they are called eigenvectors)
- such that:

$$\mathbf{M} = \sum_{i=1}^n \lambda_i \mathbf{v}_i \mathbf{v}_i^{\top}$$

Which implies: (for all i)

$$\mathbf{M} \mathbf{v}_i = \lambda_i \mathbf{v}_i$$

If we change the axes of the Euclidean space $\mathbb{R}^n$ to the eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$, then the matrix $\mathbf{M}$ becomes diagonal, with $\lambda_1, \dots, \lambda_n$ on the diagonal.

In other words, symmetric matrices stretch space independently along orthogonal directions.

Variational characterizations

- Let $\mathbf{M}$ be a real symmetric $(n \times n)$ -matrix.
 - Eigenvalues: $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$
 - Eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

Theorem:

$$\lambda_k = \min_{k\text{-dim space } \mathcal{V}} \max_{\mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\}} R_{\mathbf{M}}(\mathbf{x})$$

Rayleigh quotient of $\mathbf{x}$
with respect to $\mathbf{M}$:

$$R_{\mathbf{M}}(\mathbf{x}) = \frac{\mathbf{x}^\top \mathbf{M} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}$$

$$\mathbf{x}^\top \mathbf{L} \mathbf{x} = \sum_{\{u,v\} \in E} (x_u - x_v)^2$$

$$\mathbf{x}^\top \mathbf{N} \mathbf{x} = \frac{1}{d} \sum_{\{u,v\} \in E} (x_u - x_v)^2$$

Variational characterizations

Rayleigh quotient of $\mathbf{x}$ with respect to $\mathbf{M}$:

$$R_{\mathbf{M}}(\mathbf{x}) = \frac{\mathbf{x}^\top \mathbf{M} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}$$

- Let $\mathbf{M}$ be a real symmetric $(n \times n)$ -matrix.
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 - Eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

Theorem:

$$\lambda_k = \min_{k\text{-dim space } \mathcal{V}} \max_{\mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\}} R_{\mathbf{M}}(\mathbf{x})$$

Proof ($\geq$):

Consider $\mathcal{V} = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$

- Write $\mathbf{x} \in \mathcal{V}$ as $\mathbf{x} = \sum_{i=1}^k \alpha_i \mathbf{v}_i$.

- $R_{\mathbf{M}}(\mathbf{x}) = \frac{\sum_{i=1}^k \alpha_i^2 \lambda_i}{\sum_{i=1}^k \alpha_i^2}.$

- $\max_{\mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\}} R_{\mathbf{M}}(\mathbf{x}) = \lambda_k.$

- $\lambda_k \geq \min_{k\text{-dim space } \mathcal{V}} \max_{\mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\}} R_{\mathbf{M}}(\mathbf{x}).$

Variational characterizations

Rayleigh quotient of $\mathbf{x}$ with respect to $\mathbf{M}$:

$$R_{\mathbf{M}}(\mathbf{x}) = \frac{\mathbf{x}^\top \mathbf{M} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}$$

- Let $\mathbf{M}$ be a real symmetric $(n \times n)$ -matrix.
 - Eigenvalues: $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$
 - Eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

Theorem:

$$\lambda_k = \min_{k\text{-dim space } \mathcal{V}} \max_{\mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\}} R_{\mathbf{M}}(\mathbf{x})$$

Proof ($\leq$):

Consider any k -dimensional space $\mathcal{V}$.

- $\exists \mathbf{x} \in \mathcal{V} \cap \text{span}\{\mathbf{v}_k, \dots, \mathbf{v}_n\} \setminus \{\mathbf{0}\}$.
 - $\mathbf{x} = \sum_{i=k}^n \alpha_i \mathbf{v}_i$.
 - $R_{\mathbf{M}}(\mathbf{x}) = \frac{\sum_{i=k}^n \alpha_i^2 \lambda_i}{\sum_{i=k}^n \alpha_i^2} \geq \lambda_k$.
- $\lambda_k \leq \min_{k\text{-dim space } \mathcal{V}} \max_{\mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\}} R_{\mathbf{M}}(\mathbf{x})$.

Variational characterizations

Rayleigh quotient of $\mathbf{x}$
with respect to $\mathbf{M}$:

$$R_{\mathbf{M}}(\mathbf{x}) = \frac{\mathbf{x}^\top \mathbf{M} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}$$

- Let $\mathbf{M}$ be a real symmetric $(n \times n)$ -matrix.
 - Eigenvalues: $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$
 - Eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

Theorem:

$$\lambda_k = \min_{k\text{-dim space } \mathcal{V}} \max_{\mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\}} R_{\mathbf{M}}(\mathbf{x})$$

$$\lambda_1 = \min_{\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}} R_{\mathbf{M}}(\mathbf{x})$$

Moreover, any minimizer $\mathbf{x}$ is a corresponding eigenvector.

Consider $-\mathbf{M}$.

$$\lambda_n = \max_{\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}} R_{\mathbf{M}}(\mathbf{x})$$

Moreover, any maximizer $\mathbf{x}$ is a corresponding eigenvector.

Variational characterizations

Rayleigh quotient of $\mathbf{x}$ with respect to $\mathbf{M}$:

$$R_{\mathbf{M}}(\mathbf{x}) = \frac{\mathbf{x}^\top \mathbf{M} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}$$

- Let $\mathbf{M}$ be a real symmetric $(n \times n)$ -matrix.
 - Eigenvalues: $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$
 - Eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

Theorem:

$$\lambda_k = \min_{k\text{-dim space } \mathcal{V}} \max_{\mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\}} R_{\mathbf{M}}(\mathbf{x})$$

$$\lambda_1 = \min_{\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}} R_{\mathbf{M}}(\mathbf{x})$$

Consider $-\mathbf{M}$.

$$\lambda_n = \max_{\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}} R_{\mathbf{M}}(\mathbf{x})$$

Theorem: We omit the proof.

$$\lambda_k = \min_{\mathbf{x} \neq \mathbf{0}, \mathbf{x} \perp \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_{k-1}\}} R_{\mathbf{M}}(\mathbf{x})$$

$$\lambda_2 = \min_{\mathbf{x} \neq \mathbf{0}, \mathbf{x} \perp \mathbf{v}_1} R_{\mathbf{M}}(\mathbf{x})$$

A natural extension

Application 1: Spectral embedding

Data

Graph

Embedding the vertices to $\mathbb{R}^k$ by taking the eigenvectors corresponding to the first k non-zero eigenvalues.

We will later see that, regardless of the graph, $\mathbf{1}$ is an eigenvector of $\mathbf{N}$ (as well as $\mathbf{L}$) with eigenvalue zero. This eigenvector therefore carries no informative content and should be ignored.

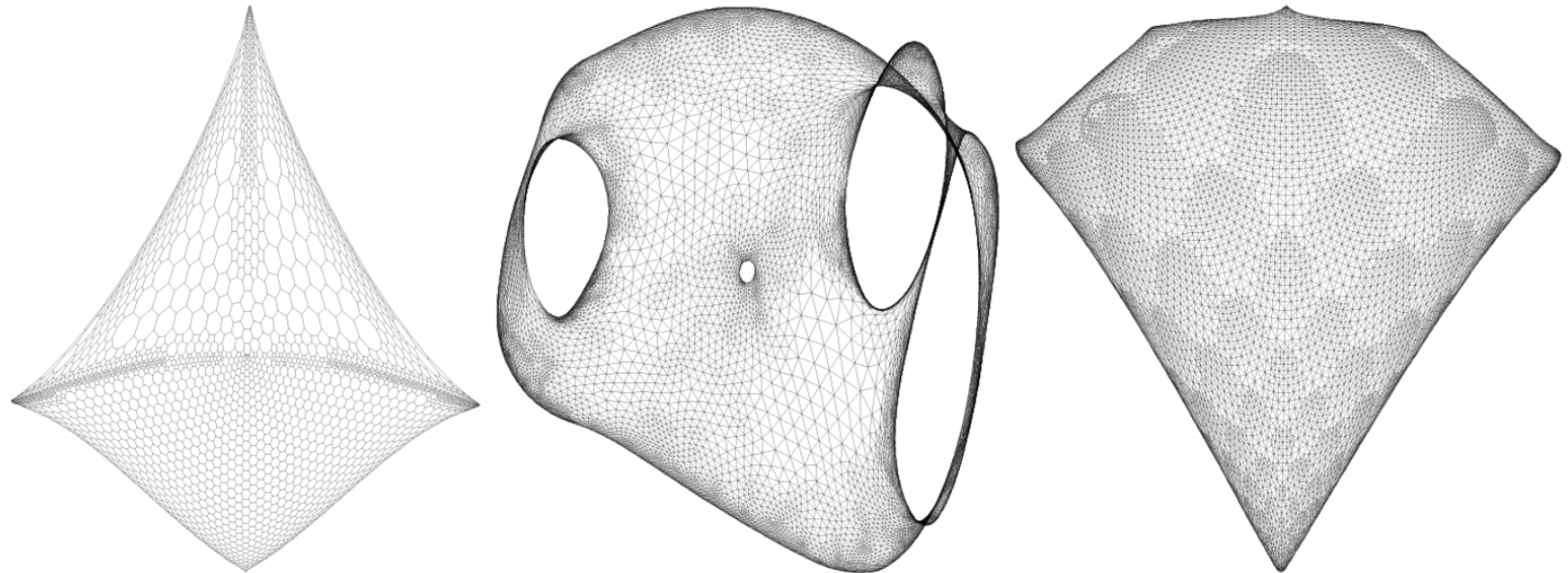
Application 1: Spectral embedding

Data

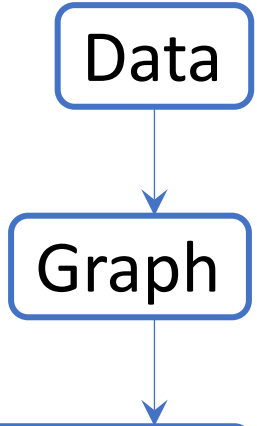
Graph

Embedding the vertices to $\mathbb{R}^k$ by taking the eigenvectors corresponding to the first k non-zero eigenvalues.

- Useful for: Minimizing $\mathbf{x}^\top \mathbf{L} \mathbf{x} = \sum_{\{u,v\} \in E} (x_u - x_v)^2$ leads to natural drawings.
- Visualization



Application 1: Spectral embedding



Embedding the vertices to $\mathbb{R}^k$ by taking the eigenvectors corresponding to the first k non-zero eigenvalues.

- Useful for:
 - Visualization
 - **Clustering + dimension reduction** https://en.wikipedia.org/wiki/Spectral_clustering
 - Spectral embedding turns high-dimensional data into points in a low-dimensional space where clusters become visible.
 - Eigenvectors can capture many features of the data that many traditional clustering methods fail to detect.



Already many theoretical evidences

The eigenvector equation

$$Ax = \lambda x$$

Application 2: Network analysis

- Intuitively, an eigenvector x captures the influence of nodes in a network:

Connections to influential nodes enhance your own influence.

$$x_v = \frac{1}{\lambda} \sum_{u \in N(v)} x_u$$

Some applications:

- Eigenvector centrality is the unique measure satisfying certain natural axioms for a **ranking system**.
- In **neuroscience**, the eigenvector centrality of a neuron in a model neural network has been found to correlate with its relative firing rate.
- Eigenvector centrality and related concepts have been used to model **opinion influence in sociology and economics**.
- **Google's PageRank** is based on a variant of Eigenvector centrality.

https://en.wikipedia.org/wiki/Eigenvector_centrality

The smallest eigenvalue λ_1

- Consider the normalized Laplacian N of a graph $G = (V, E)$.
 - Eigenvalues: $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$

Theorem:

- $\lambda_1 = 0$, with $\mathbf{1}$ being a corresponding eigenvector.

Rayleigh quotient of $\mathbf{x}$ with respect to N :

$$R_N(\mathbf{x}) = \frac{\mathbf{x}^\top N \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} = \frac{\sum_{\{u,v\} \in E} (x_u - x_v)^2}{d \sum_{v \in V} x_v^2} \geq 0$$

$$\lambda_1 \geq 0$$

$$\lambda_1 = \min_{\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}} R_N(\mathbf{x}) \leq R_N(\mathbf{1}) = \frac{\sum_{\{u,v\} \in E} (1 - 1)^2}{d \sum_{v \in V} 1^2} = 0$$

$$\lambda_1 \leq 0$$

The eigenvalue λ_k

- Consider the normalized Laplacian N of a graph $G = (V, E)$.
 - Eigenvalues: $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$

Theorem:

- $\lambda_k = 0$ if and only if G has at least k connected components.

$$R_N(\mathbf{x}) = \frac{\mathbf{x}^\top N \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} = \frac{\sum_{\{u,v\} \in E} (x_u - x_v)^2}{d \sum_{v \in V} x_v^2} = 0$$

if and only if

For every connected component S ,
 $x_u = x_v \quad \forall u, v \in S$

G has at least k connected components $S_1, \dots, S_k$.

Set $\mathcal{V} = \text{span}\{\mathbf{1}_{S_1}, \dots, \mathbf{1}_{S_k}\}$.

- $\forall \mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\}, R_N(\mathbf{x}) = 0$

$$\lambda_k = \min_{k\text{-dim space } \mathcal{V}} \max_{\mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\}} R_N(\mathbf{x})$$

$$\lambda_k = 0$$

The eigenvalue λ_k

- Consider the normalized Laplacian N of a graph $G = (V, E)$.
 - Eigenvalues: $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$

Theorem:

- $\lambda_k = 0$ if and only if G has at least k connected components.

$$R_N(\mathbf{x}) = \frac{\mathbf{x}^\top N \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} = \frac{\sum_{\{u,v\} \in E} (x_u - x_v)^2}{d \sum_{v \in V} x_v^2} = 0$$

if and only if

For every connected component S ,
 $x_u = x_v \quad \forall u, v \in S$

$k = \text{dimension of } \mathcal{V} \leq \text{the number of connected components.}$

$$\lambda_k = 0$$

There is a k -dimensional subspace $\mathcal{V}$ of $\mathbb{R}^n$ such that:

- $\forall \mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\}, R_N(\mathbf{x}) = 0$

The value of $\mathbf{x}$ is constant within each connected component.

The largest eigenvalue λ_n

- Consider the normalized Laplacian N of a graph $G = (V, E)$.
 - Eigenvalues: $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$

Theorem:

- $\lambda_n \leq 2$.
- $\lambda_n = 2$ if and only if G has a bipartite connected component.

$$\lambda_n = \max_{x \in \mathbb{R}^n \setminus \{0\}} R_N(x) = 2 - \min_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\sum_{\{u,v\} \in E} (x_u + x_v)^2}{d \sum_{v \in V} x_v^2} \leq 2$$

$$\begin{aligned} R_N(x) &= \frac{\sum_{\{u,v\} \in E} (x_u - x_v)^2}{d \sum_{v \in V} x_v^2} \\ &= \frac{2 \sum_{\{u,v\} \in E} (x_u^2 + x_v^2) - \sum_{\{u,v\} \in E} (x_u + x_v)^2}{d \sum_{v \in V} x_v^2} \\ &= 2 - \frac{\sum_{\{u,v\} \in E} (x_u + x_v)^2}{d \sum_{v \in V} x_v^2} \end{aligned}$$

The largest eigenvalue λ_n

- Consider the normalized Laplacian N of a graph $G = (V, E)$.
 - Eigenvalues: $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$

Theorem:

- $\lambda_n \leq 2$.
- $\lambda_n = 2$ if and only if G has a bipartite connected component.

$$\lambda_n = 2 - \min_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\sum_{\{u,v\} \in E} (x_u + x_v)^2}{d \sum_{v \in V} x_v^2}$$

- Suppose G has a bipartite connected component S with bipartition $S = A \cup B$.
- Setting $x = \mathbf{1}_A - \mathbf{1}_B$ makes $\frac{\sum_{\{u,v\} \in E} (x_u + x_v)^2}{d \sum_{v \in V} x_v^2} = 0$.
- Therefore, $\lambda_n = 2$.

The largest eigenvalue λ_n

- Consider the normalized Laplacian N of a graph $G = (V, E)$.
 - Eigenvalues: $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$

Theorem:

- $\lambda_n \leq 2$.
- $\lambda_n = 2$ if and only if G has a bipartite connected component.

$$\lambda_n = 2 - \min_{x \in \mathbb{R}^n \setminus \{0\}} \frac{\sum_{\{u,v\} \in E} (x_u + x_v)^2}{d \sum_{v \in V} x_v^2}$$

- Suppose $\lambda_n = 2$.
- There exists $x \in \mathbb{R}^n \setminus \{0\}$ such that $\frac{\sum_{\{u,v\} \in E} (x_u + x_v)^2}{d \sum_{v \in V} x_v^2} = 0$.
- Define:
 - $A = \{v \in V \mid x_v > 0\} \neq \emptyset$
 - $B = \{v \in V \mid x_v < 0\} \neq \emptyset$
- $A \cup B$ is a union of bipartite connected components.

Outlook

- **Next:** We will show that:

$$\frac{\lambda_2}{2} \leq \Phi(G) \leq \sqrt{2\lambda_2} \quad \text{Cheeger's inequality}$$

where λ_2 is the second eigenvalue of the normalized Laplacian N of G .

Moreover, given the eigenvector v_2 , we can obtain a cut of conductance at most $\sqrt{2\lambda_2}$ in polynomial time.

- **Next:** Efficient eigenvector computation.

Useful in many applications, both in practice and in theory.

λ_2 and conductance

- **Recall:** $\lambda_2 = 0$ if and only if G has at least 2 connected components.



The second eigenvalue of the normalized Laplacian N of G .

- Intuitively, there should be a robust version of this fact:
 - λ_2 is small if and only if G has a sparse cut.

- We will prove this:

$$\text{Cheeger inequality: } \frac{\lambda_2}{2} \leq \Phi(G) \leq \sqrt{2\lambda_2}$$

Part 1: $\frac{\lambda_2}{2} \leq \Phi(G)$

Let $(S, V \setminus S)$ be a sparsest cut: $\Phi(S) = \Phi(G)$.

$$R_N(\mathbf{1}_S) = \frac{|E(S, V \setminus S)|}{\text{vol}(S)}$$

$$R_N(\mathbf{1}_{V \setminus S}) = \frac{|E(S, V \setminus S)|}{\text{vol}(V \setminus S)}$$

Rayleigh quotient of $\mathbf{x}$ with respect to $\mathbf{N}$:

$$R_N(\mathbf{x}) = \frac{\mathbf{x}^\top \mathbf{N} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} = \frac{\sum_{\{u,v\} \in E} (x_u - x_v)^2}{d \sum_{v \in V} x_v^2}$$

$$\begin{aligned} \lambda_2 &= \min_{2\text{-dim space } \mathcal{V}} \max_{\mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\}} R_N(\mathbf{x}) \\ &\leq \max_{\mathbf{z} \in \text{span}(\mathbf{1}_S, \mathbf{1}_{V \setminus S}) \setminus \{\mathbf{0}\}} R_N(\mathbf{z}) \\ &\leq 2 \max\{R_N(\mathbf{1}_S), R_N(\mathbf{1}_{V \setminus S})\} \\ &= 2 \cdot \frac{|E(S, V \setminus S)|}{\min\{\text{vol}(S), \text{vol}(V \setminus S)\}} \\ &= 2 \cdot \Phi(S) \\ &= 2 \cdot \Phi(G) \end{aligned}$$

Recall: $\lambda_2 = \min_{2\text{-dim space } \mathcal{V}} \max_{\mathbf{x} \in \mathcal{V} \setminus \{\mathbf{0}\}} R_N(\mathbf{x})$

Lemma:

- Let $\mathbf{x}$ and $\mathbf{y}$ be two orthogonal vectors.
- Let $\mathbf{z} \in \text{span}(\mathbf{x}, \mathbf{y}) \setminus \{\mathbf{0}\}$.

$$R_N(\mathbf{z}) \leq 2 \max\{R_N(\mathbf{x}), R_N(\mathbf{y})\}$$

Part 1: $\frac{\lambda_2}{2} \leq \Phi(G)$

Proof of the lemma:

- It suffices to prove it for the case of $\mathbf{z} = \mathbf{x} + \mathbf{y}$.

$$R_M(\alpha \mathbf{x} + \beta \mathbf{y}) \leq 2 \max\{R_M(\alpha \mathbf{x}), R_M(\beta \mathbf{y})\} = 2 \max\{R_M(\mathbf{x}), R_M(\mathbf{y})\}$$

Lemma:

- Let $\mathbf{x}$ and $\mathbf{y}$ be two orthogonal vectors.
- Let $\mathbf{z} \in \text{span}(\mathbf{x}, \mathbf{y}) \setminus \{\mathbf{0}\}$.

$$R_N(\mathbf{z}) \leq 2 \max\{R_N(\mathbf{x}), R_N(\mathbf{y})\}$$

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N is a real symmetric $(n \times n)$ -matrix:

- Eigenvalues: $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$
- Eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

We write:

- $\mathbf{x} = \sum_{i=1}^n a_i \mathbf{v}_i$
- $\mathbf{y} = \sum_{i=1}^n b_i \mathbf{v}_i$

$$\begin{aligned} R_N(\mathbf{z}) &= \frac{\mathbf{z}^\top N \mathbf{z}}{\mathbf{z}^\top \mathbf{z}} \\ &= \frac{\sum_{i=1}^n \lambda_i (a_i + b_i)^2}{\|\mathbf{x} + \mathbf{y}\|^2} \\ &\leq \frac{2 \sum_{i=1}^n \lambda_i (a_i^2 + b_i^2)}{\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2} \end{aligned}$$

$$\mathbf{x} \perp \mathbf{y} \Rightarrow \|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$$

$$(a_i + b_i)^2 \leq 2a_i^2 + 2b_i^2$$

Lemma:

- Let $\mathbf{x}$ and $\mathbf{y}$ be two orthogonal vectors.
- Let $\mathbf{z} \in \text{span}(\mathbf{x}, \mathbf{y}) \setminus \{\mathbf{0}\}$.

$$R_N(\mathbf{z}) \leq 2 \max\{R_N(\mathbf{x}), R_N(\mathbf{y})\}$$

Part 1: $\frac{\lambda_2}{2} \leq \Phi(G)$

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- Eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

We write:

- $\mathbf{x} = \sum_{i=1}^n a_i \mathbf{v}_i$
- $\mathbf{y} = \sum_{i=1}^n b_i \mathbf{v}_i$

$$\begin{aligned} R_N(\mathbf{z}) &= \frac{\mathbf{z}^\top N \mathbf{z}}{\mathbf{z}^\top \mathbf{z}} \\ &= \frac{\sum_{i=1}^n \lambda_i (a_i + b_i)^2}{\|\mathbf{x} + \mathbf{y}\|^2} \\ &\leq \frac{2 \sum_{i=1}^n \lambda_i (a_i^2 + b_i^2)}{\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2} \\ &= \frac{2(\mathbf{x}^\top N \mathbf{x} + \mathbf{y}^\top N \mathbf{y})}{\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2} \\ &\leq 2 \max\{R_N(\mathbf{x}), R_N(\mathbf{y})\} \end{aligned}$$

Lemma:

- Let $\mathbf{x}$ and $\mathbf{y}$ be two orthogonal vectors.
- Let $\mathbf{z} \in \text{span}(\mathbf{x}, \mathbf{y}) \setminus \{\mathbf{0}\}$.

$$R_N(\mathbf{z}) \leq 2 \max\{R_N(\mathbf{x}), R_N(\mathbf{y})\}$$

Part 2: $\Phi(G) \leq \sqrt{2\lambda_2}$

Lemma:

- Let $\mathbf{x}$ be any vector orthogonal to $\mathbf{1}$.
- There is a polynomial-time algorithm that computes a cut with conductance at most $\sqrt{2R_N(\mathbf{x})}$.

Setting $\mathbf{x} = \mathbf{v}_2$ yields a cut with conductance at most $\sqrt{2\lambda_2}$.

$$\Phi(G) \leq \sqrt{2\lambda_2}$$

Part 2: $\Phi(G) \leq \sqrt{2\lambda_2}$

Lemma:

- Let $\mathbf{x}$ be any vector orthogonal to $\mathbf{1}$. This yields a cut with conductance at most $\sqrt{2 \min\{R_N(\mathbf{x}^+), R_N(\mathbf{x}^-)\}} \leq \sqrt{2R_N(\mathbf{x})}$.
- There is a polynomial-time algorithm that computes a cut with conductance at most $\sqrt{2R_N(\mathbf{x})}$.

Proof of the lemma:

- $\mathbf{x}' = \mathbf{x} - m\mathbf{1}$
 - m is the median of $\mathbf{x}$.
- $\mathbf{x}' = \mathbf{x}^+ - \mathbf{x}^-$
 - $x'_v = \begin{cases} x_v & \text{if } x_v > 0 \\ 0 & \text{otherwise} \end{cases}$
 - $x'_v = \begin{cases} -x_v & \text{if } x_v < 0 \\ 0 & \text{otherwise} \end{cases}$

At least half of the vertices have zero values.

Setting $\mathbf{y}$ to be $\mathbf{x}^+$ or $\mathbf{x}^-$

Claim 1:

$$R_N(\mathbf{x}) \geq R_N(\mathbf{x}') \geq \min\{R_N(\mathbf{x}^+), R_N(\mathbf{x}^-)\}$$

Claim 2:

Let $\mathbf{y} \in \mathbb{R}_{\geq 0}^n$. There exists a threshold $t > 0$ such that $S_t = \{v : y_v \geq t\}$ satisfies:

$$\frac{|E(S_t, V \setminus S_t)|}{d|S_t|} \leq \sqrt{2R_N(\mathbf{y})}$$

$$\Phi(S_t) = \frac{|E(S_t, V \setminus S_t)|}{d \min\{|S_t|, |V \setminus S_t|\}} = \frac{|E(S_t, V \setminus S_t)|}{d|S_t|} \leq \sqrt{2R_N(\mathbf{y})}$$

Proof of Claim 1

Lemma:

- Let $\mathbf{x}$ be any vector orthogonal to $\mathbf{1}$.
- There is a polynomial-time algorithm that computes a cut with conductance at most $\sqrt{2R_N(\mathbf{x})}$.

Claim 1:

$$R_N(\mathbf{x}) \geq R_N(\mathbf{x}') \geq \min\{R_N(\mathbf{x}^+), R_N(\mathbf{x}^-)\}$$

- $\mathbf{x}' = \mathbf{x} - m\mathbf{1}$
 - m is the median of $\mathbf{x}$.

$$R_N(\mathbf{x}') = \frac{(\mathbf{x} - m\mathbf{1})^\top N(\mathbf{x} - m\mathbf{1})}{\|\mathbf{x} - m\mathbf{1}\|^2} = \frac{\mathbf{x}^\top N\mathbf{x}}{\|\mathbf{x} - m\mathbf{1}\|^2} = \frac{\mathbf{x}^\top N\mathbf{x}}{\|\mathbf{x}\|^2 + m^2} \leq \frac{\mathbf{x}^\top N\mathbf{x}}{\|\mathbf{x}\|^2} = R_N(\mathbf{x})$$

$\mathbf{1}$ is an eigenvector of N with a zero eigenvalue.

$\mathbf{x} \perp \mathbf{1}$

Proof of Claim 1

Lemma:

- Let $\mathbf{x}$ be any vector orthogonal to $\mathbf{1}$.
- There is a polynomial-time algorithm that computes a cut with conductance at most $\sqrt{2R_N(\mathbf{x})}$.

- $\mathbf{x}' = \mathbf{x}^+ - \mathbf{x}^-$
 - $x'_v = \begin{cases} x_v^+ & \text{if } x_v^+ > 0 \\ 0 & \text{otherwise} \end{cases}$
 - $x'_v = \begin{cases} -x_v^- & \text{if } x_v^- < 0 \\ 0 & \text{otherwise} \end{cases}$

Claim 1:

$$R_N(\mathbf{x}) \geq R_N(\mathbf{x}') \geq \min\{R_N(\mathbf{x}^+), R_N(\mathbf{x}^-)\}$$

$$R_N(\mathbf{x}') = \frac{\sum_{\{u,v\} \in E} (x_u - x_v)^2}{d \|\mathbf{x}'\|^2} \geq \frac{\sum_{\{u,v\} \in E} ((x_u^+ - x_v^+)^2 + (x_u^- - x_v^-)^2)}{d(\|\mathbf{x}^+\|^2 + \|\mathbf{x}^-\|^2)} \geq \min\{R_N(\mathbf{x}^+), R_N(\mathbf{x}^-)\}$$

$$\begin{aligned} \|\mathbf{x}'\|^2 &= \|\mathbf{x}^+\|^2 + \|\mathbf{x}^-\|^2 \\ (x_u - x_v)^2 &\geq (x_u^+ - x_v^+)^2 + (x_u^- - x_v^-)^2 \end{aligned}$$

Proof of Claim 2

- Assume $\max_v y_v = 1$. Multiplication by a scalar does not affect the Rayleigh quotient of a vector.
- Select t randomly in such a way that t^2 is uniformly distributed in $[0,1]$.
- Consider $S_t = \{v : y_v \geq t\}$.

We will prove:

$$\frac{\mathbb{E}[|E(S_t, V \setminus S_t)|]}{d\mathbb{E}[|S_t|]} \leq \sqrt{2R_N(\mathbf{y})}$$

Claim 2:

Let $\mathbf{y} \in \mathbb{R}_{\geq 0}^n$. There exists a threshold $t > 0$ such that $S_t = \{v : y_v \geq t\}$ satisfies:

$$\frac{|E(S_t, V \setminus S_t)|}{d|S_t|} \leq \sqrt{2R_N(\mathbf{y})}$$

Proof of Claim 2

$$\text{Sub-claim 1: } d\mathbb{E}[|S_t|] = d \sum_v y_v^2$$

$$\text{Sub-claim 2: } \mathbb{E}[|E(S_t, V \setminus S_t)|] = \sqrt{2 \left(\sum_{\{u,v\} \in E} (y_u - y_v)^2 \right) (d \sum_v y_v^2)}$$

- Assume $\max_v y_v = 1$.
- Select t randomly in such a way that t^2 is uniformly distributed in $[0,1]$.
- Consider $S_t = \{v : y_v \geq t\}$.

$$\frac{\mathbb{E}[|E(S_t, V \setminus S_t)|]}{d\mathbb{E}[|S_t|]} \leq \sqrt{2 \cdot \frac{\sum_{\{u,v\} \in E} (y_u - y_v)^2}{d \sum_{v \in V} y_v^2}} = \sqrt{2R_N(\mathbf{y})}$$

We will prove:

$$\frac{\mathbb{E}[|E(S_t, V \setminus S_t)|]}{d\mathbb{E}[|S_t|]} \leq \sqrt{2R_N(\mathbf{y})}$$

Proof of Claim 2

$$\text{Sub-claim 1: } d\mathbb{E}[|S_t|] = d \sum_v y_v^2$$

$$\text{Sub-claim 2: } \mathbb{E}[|E(S_t, V \setminus S_t)|] = \sqrt{2 \left(\sum_{\{u,v\} \in E} (y_u - y_v)^2 \right) (d \sum_v y_v^2)}$$

- Assume $\max_v y_v = 1$.
- Select t randomly in such a way that t^2 is uniformly distributed in $[0,1]$.
- Consider $S_t = \{v : y_v \geq t\}$.

Proof of Sub-claim 1:

For each vertex v , it is added to S_t when $t^2 \in [0, y_v^2]$, which happens with probability y_v^2 .

Linearity of expectations:

$$\mathbb{E}[|S_t|] = \sum_v y_v^2$$

$$\text{Sub-claim 1: } d\mathbb{E}[|S_t|] = d \sum_v y_v^2$$

Proof of Claim 2

$$\text{Sub-claim 2: } \mathbb{E}[|E(S_t, V \setminus S_t)|] = \sqrt{2 \left(\sum_{\{u,v\} \in E} (y_u - y_v)^2 \right) (d \sum_v y_v^2)}$$

- Assume $\max_v y_v = 1$.
- Select t randomly in such a way that t^2 is uniformly distributed in $[0,1]$.
- Consider $S_t = \{v : y_v \geq t\}$.

Proof of Sub-claim 2:

For each edge $e = \{u, v\}$, it belongs to $E(S_t, V \setminus S_t)$ when

$$t^2 \in (\min\{y_u^2, y_v^2\}, \max\{y_u^2, y_v^2\}],$$

which happens with probability

$$|y_u^2 - y_v^2|.$$

$$\mathbb{E}[|E(S_t, V \setminus S_t)|] = \sum_{\{u,v\} \in E} |y_u^2 - y_v^2|$$

$$= \sum_{\{u,v\} \in E} (y_u + y_v) |y_u - y_v|$$

$$\leq \underbrace{\sqrt{\sum_{\{u,v\} \in E} (y_u + y_v)^2}}_{\sum_{\{u,v\} \in E} (y_u + y_v)^2 \leq 2 \sum_{\{u,v\} \in E} (y_u^2 + y_v^2) = 2d \sum_v y_v^2} \sqrt{\sum_{\{u,v\} \in E} (y_u - y_v)^2}$$

$$\sum_{\{u,v\} \in E} (y_u + y_v)^2 \leq 2 \sum_{\{u,v\} \in E} (y_u^2 + y_v^2) = 2d \sum_v y_v^2$$

Cheeger vs. Leighton–Rao

- Cheeger inequality yields more practical algorithms.

Small and precise constants:

$$\frac{\lambda_2}{2} \leq \Phi(G) \leq \sqrt{2\lambda_2}$$

Simple and fast algorithm:

After sorting the vertices according to a given vector $\mathbf{x}$ orthogonal to $\mathbf{1}$, finding a cut with conductance at most $\sqrt{2R_N(\mathbf{x})}$ can be done in **linear time**. ← Exercise!

Setting $\mathbf{x} = \mathbf{v}_2$ yields a cut with conductance at most $\sqrt{2\lambda_2} \leq 2\sqrt{\Phi(G)}$.

$$\frac{\lambda_2}{2} \leq \Phi(G)$$

Extension 1: multi-way cuts

- **Recall:** $\lambda_k = 0$ if and only if G has at least k connected components.
- Similar to Cheeger inequality, there is a robust version of this fact.

$$\Phi_k(G) = \min_{\text{disjoint } S_1 \subseteq V, S_2 \subseteq V, \dots, S_k \subseteq V} \max_{1 \leq i \leq k} \frac{|E(S_i, V \setminus S_i)|}{d|S_i|}$$

$$\frac{\lambda_k}{2} \leq \Phi_k(G) \leq O(k^2) \cdot \lambda_k$$

<https://doi.org/10.1145/2665063>

Extension 2: bipartiteness

- **Recall:** $\lambda_n = 2$ if and only if G has a bipartite connected component.
- Similar to Cheeger inequality, there is a robust version of this fact.

A graph $G = (V, E)$ is **ϵ -close to bipartite** if one can make it bipartite by removing at most an ϵ -fraction of its edges.

Theorem: If A graph $G = (V, E)$ is ϵ -close to bipartite, then $\lambda_n \geq 2(1 - \epsilon)$.

Proof:

- Suppose $G = (V, E)$ is ϵ -close to a bipartite graph with the bipartition $V = A \cup B$.
- Let $\mathbf{y} = \mathbf{1}_A - \mathbf{1}_B$.

$$2 - \lambda_n = \min_{\mathbf{x} \in \mathbb{R}^n \setminus \{\mathbf{0}\}} \frac{\sum_{\{u,v\} \in E} (x_u + x_v)^2}{d \sum_{v \in V} x_v^2} \leq \frac{\sum_{\{u,v\} \in E} (y_u + y_v)^2}{d \sum_{v \in V} y_v^2} \leq \frac{4\epsilon|E|}{2|E|} = 2\epsilon$$

Extension 2: bipartiteness

- **Recall:** $\lambda_n = 2$ if and only if G has a bipartite connected component.
- Similar to Cheeger inequality, there is a robust version of this fact.

A graph $G = (V, E)$ is **ϵ -close to bipartite** if one can make it bipartite by removing at most an ϵ -fraction of its edges.

Theorem: If A graph $G = (V, E)$ is ϵ -close to bipartite, then $\lambda_n \geq 2(1 - \epsilon)$.

Moreover, using approximate eigenvector computation, a set $D \subseteq E$ with $|D| \in O(\sqrt{\epsilon}) \cdot |E|$ such that $G' = (V, E \setminus D)$ is bipartite can be computed in polynomial time.

Recap

$$\text{Cheeger inequality: } \frac{\lambda_2}{2} \leq \Phi(G) \leq \sqrt{2\lambda_2}$$

Lemma:

- Let $\mathbf{x}$ be any vector orthogonal to $\mathbf{1}$.
- There is a polynomial-time algorithm that computes a cut with conductance at most $\sqrt{2R_N(\mathbf{x})}$.



Setting $\mathbf{x} = \mathbf{v}_2$ yields a cut with conductance at most $\sqrt{2\lambda_2}$.

Our goal: Obtaining this in polynomial time.

From any vector $\mathbf{x} \perp \mathbf{1}$ with $R_N(\mathbf{x}) \leq \lambda_2 + \epsilon$, we get a cut with conductance at most $\sqrt{2(\lambda_2 + \epsilon)}$.

Setting

- Let $\mathbf{M}$ be a real symmetric $(n \times n)$ -matrix.
 - Eigenvalues: $0 \leq \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$
 - Eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ (orthonormal vectors)

Eigenvalue range for d -regular graphs:

- $\mathbf{A}$ (adjacency): $[-d, d]$ ✗
- $\mathbf{L} = d\mathbf{I} - \mathbf{A}$ (Laplacian): $[0, 2d]$ ✓
- $\mathbf{N} = \mathbf{I} - \frac{1}{d}\mathbf{A}$ (normalized Laplacian): $[0, 2]$ ✓

Setting

- Let M be a real symmetric $(n \times n)$ -matrix.
 - Eigenvalues: $0 \leq \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$
 - Eigenvectors $v_1, v_2, \dots, v_n$ (orthonormal vectors)
- **First goal:** Approximating the largest eigenvector v_n .
 - Find a vector x with $R_M(x) \geq (1 - \epsilon) \cdot \lambda_n$.



We will later extend this to:

- The second largest eigenvector v_{n-1} .
- The smallest eigenvector v_1 .
- The second smallest eigenvector v_2 (our main goal).

The power method

Algorithm **POWER**:

Input: a matrix $\mathbf{M}$ and a parameter k .

- Pick $\mathbf{x}_0 \in \{-1, 1\}^n$ uniformly at random.
- Return $\mathbf{x} = \mathbf{M}^k \mathbf{x}_0$.

Claim: There is a choice of parameter $k \in O\left(\frac{\log n}{\epsilon}\right)$ such that $\Pr[R_{\mathbf{M}}(\mathbf{x}) \geq (1 - \epsilon) \cdot \lambda_n] \geq \frac{3}{16}$.

Proof sketch:

The matrix $\mathbf{M}^k$ has:

- Eigenvalues: $0 \leq \lambda_1^k \leq \lambda_2^k \leq \dots \leq \lambda_n^k$
- The same eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

Suppose $\lambda_{i^*} \leq (1 - \epsilon) \cdot \lambda_n < \lambda_{i^*+1}$.

Then $\lambda_{i^*}^k \leq (1 - \epsilon)^k \cdot \lambda_n^k$

$\downarrow$
 $1/\text{poly}(n)$

To make this argument work, we need to start with a sufficiently large $c_n = \langle \mathbf{x}_0, \mathbf{v}_n \rangle$.

$$\mathbf{x}_0 = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_i \mathbf{v}_i + c_{i+1} \mathbf{v}_{i+1} + \dots + \mathbf{c}_n \mathbf{v}_n$$

$$\mathbf{x} = \mathbf{M}^k \mathbf{x}_0 = \underbrace{c_1 \lambda_1^k \mathbf{v}_1 + c_2 \lambda_2^k \mathbf{v}_2 + \dots + c_{i^*} \lambda_{i^*}^k \mathbf{v}_{i^*}}_{\text{Negligibly small}} + \underbrace{c_{i^*+1} \lambda_{i^*+1}^k \mathbf{v}_{i^*+1} + \dots + c_n \lambda_n^k \mathbf{v}_n}_{R_{\mathbf{M}}(\cdot) \geq (1 - \epsilon) \cdot \lambda_n}$$

The power method

Claim: for any unit vector $\mathbf{v}$, $\Pr \left[|\langle \mathbf{x}_0, \mathbf{v} \rangle| \geq \frac{1}{2} \right] \geq \frac{3}{16}$

Algorithm **POWER**:

Input: a matrix $\mathbf{M}$ and a parameter k .

- Pick $\mathbf{x}_0 \in \{-1, 1\}^n$ uniformly at random.
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To make this argument work, we need to start with a sufficiently large $c_n = \langle \mathbf{x}_0, \mathbf{v}_n \rangle$.

$$\mathbf{x}_0 = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_i \mathbf{v}_i + c_{i+1} \mathbf{v}_{i+1} + \cdots + \mathbf{c}_n \mathbf{v}_n$$

The power method

Claim: for any unit vector $\mathbf{v}$, $\Pr \left[|\langle \mathbf{x}_0, \mathbf{v} \rangle| \geq \frac{1}{2} \right] \geq \frac{3}{16}$

Algorithm **POWER**:

Input: a matrix $\mathbf{M}$ and a parameter k .

- Pick $\mathbf{x}_0 \in \{-1, 1\}^n$ uniformly at random.
- Return $\mathbf{x} = \mathbf{M}^k \mathbf{x}_0$.

Proof:

- $\mathbf{x}_0 = (x_1, x_2, \dots, x_n)$
- $\mathbf{v} = (v_1, v_2, \dots, v_n)$
- $Z = \langle \mathbf{x}_0, \mathbf{v} \rangle = \sum_{i=1}^n x_i v_i$

$$\mathbb{E}[Z] = \sum_{i=1}^n \mathbb{E}[x_i] v_i = 0$$

$$\mathbb{E}[Z^2] = \sum_{i=1}^n \sum_{j=1}^n \mathbb{E}[x_i x_j] v_i v_j = \sum_{i=1}^n \mathbb{E}[x_i^2] v_i^2 = \sum_{i=1}^n v_i^2 = 1$$

$$\mathbb{E}[x_i x_j] = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

$\mathbf{v}$ is a unit vector.

The power method

Claim: for any unit vector $\mathbf{v}$, $\Pr \left[|\langle \mathbf{x}_0, \mathbf{v} \rangle| \geq \frac{1}{2} \right] \geq \frac{3}{16}$

Algorithm **POWER**:

Input: a matrix $\mathbf{M}$ and a parameter k .

- Pick $\mathbf{x}_0 \in \{-1, 1\}^n$ uniformly at random.
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Proof:

- $\mathbf{x}_0 = (x_1, x_2, \dots, x_n)$
- $\mathbf{v} = (v_1, v_2, \dots, v_n)$
- $Z = \langle \mathbf{x}_0, \mathbf{v} \rangle = \sum_{i=1}^n x_i v_i$

$$\mathbb{E}[Z] = \sum_{i=1}^n \mathbb{E}[x_i] v_i = 0$$

$$\mathbb{E}[Z^2] = \sum_{i=1}^n \sum_{j=1}^n \mathbb{E}[x_i x_j] v_i v_j = \sum_{i=1}^n \mathbb{E}[x_i^2] v_i^2 = \sum_{i=1}^n v_i^2 = 1$$

$$\mathbb{E}[Z^4] = \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \sum_{\ell=1}^n \underbrace{\mathbb{E}[x_i x_j x_k x_\ell]}_{\text{bracket}} v_i v_j v_k v_\ell = 3 \sum_{i=1}^n \mathbb{E}[x_i^2] v_i^2 \sum_{j=1}^n \mathbb{E}[x_j^2] v_j^2 - 2 \sum_{i=1}^n \mathbb{E}[x_i^4] v_i^4$$

$$i \notin \{j, k, \ell\} \rightarrow \mathbb{E}[x_i x_j x_k x_\ell] = 0$$

$$j \notin \{i, k, \ell\} \rightarrow \mathbb{E}[x_i x_j x_k x_\ell] = 0$$

$$k \notin \{i, j, \ell\} \rightarrow \mathbb{E}[x_i x_j x_k x_\ell] = 0$$

$$\ell \notin \{i, j, k\} \rightarrow \mathbb{E}[x_i x_j x_k x_\ell] = 0$$

For any $i \neq j$, $\mathbb{E}[x_i^2 x_j^2]$ appears $\binom{4}{2} = 6$ times.

For any i , $\mathbb{E}[x_i^4]$ appears once.

The power method

Claim: for any unit vector $\mathbf{v}$, $\Pr \left[|\langle \mathbf{x}_0, \mathbf{v} \rangle| \geq \frac{1}{2} \right] \geq \frac{3}{16}$

Algorithm **POWER**:

Input: a matrix $\mathbf{M}$ and a parameter k .

- Pick $\mathbf{x}_0 \in \{-1, 1\}^n$ uniformly at random.
- Return $\mathbf{x} = \mathbf{M}^k \mathbf{x}_0$.

Proof:

- $\mathbf{x}_0 = (x_1, x_2, \dots, x_n)$
- $\mathbf{v} = (v_1, v_2, \dots, v_n)$
- $Z = \langle \mathbf{x}_0, \mathbf{v} \rangle = \sum_{i=1}^n x_i v_i$

$$\mathbb{E}[Z] = \sum_{i=1}^n \mathbb{E}[x_i] v_i = 0$$

$$\mathbb{E}[Z^2] = \sum_{i=1}^n \sum_{j=1}^n \mathbb{E}[x_i x_j] v_i v_j = \sum_{i=1}^n \mathbb{E}[x_i^2] v_i^2 = \sum_{i=1}^n v_i^2 = 1$$

$$\mathbb{E}[Z^4] = \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n \sum_{\ell=1}^n \mathbb{E}[x_i x_j x_k x_\ell] v_i v_j v_k v_\ell = 3 \underbrace{\sum_{i=1}^n \mathbb{E}[x_i^2] v_i^2}_{=1} \underbrace{\sum_{j=1}^n \mathbb{E}[x_j^2] v_j^2}_{=1} - 2 \underbrace{\sum_{i=1}^n \mathbb{E}[x_i^4] v_i^4}_{= \sum_{i=1}^n v_i^4 \geq 0} \leq 3$$

The power method

Claim: for any unit vector $\mathbf{v}$, $\Pr\left[|\langle \mathbf{x}_0, \mathbf{v} \rangle| \geq \frac{1}{2}\right] \geq \frac{3}{16}$

Algorithm **POWER**:

Input: a matrix $\mathbf{M}$ and a parameter k .

- Pick $\mathbf{x}_0 \in \{-1, 1\}^n$ uniformly at random.
- Return $\mathbf{x} = \mathbf{M}^k \mathbf{x}_0$.

Proof:

- $\mathbf{x}_0 = (x_1, x_2, \dots, x_n)$
- $\mathbf{v} = (v_1, v_2, \dots, v_n)$
- $Z = \langle \mathbf{x}_0, \mathbf{v} \rangle = \sum_{i=1}^n x_i v_i$

- $\mathbb{E}[Z] = 0$
- $\mathbb{E}[Z^2] = 1$
- $\mathbb{E}[Z^4] \leq 3$
- $\text{Var}[Z^2] = \mathbb{E}[Z^4] - \mathbb{E}[Z^2]^2 < \infty$

$$\begin{aligned}\Pr\left[|\langle \mathbf{x}_0, \mathbf{v} \rangle| \geq \frac{1}{2}\right] &= \Pr\left[|Z| \geq \frac{1}{2}\right] = \Pr\left[Z^2 \geq \frac{1}{4}\right] \\ &= \Pr[Z^2 \geq \delta \mathbb{E}[Z^2]] \geq (1 - \delta)^2 \frac{\mathbb{E}[Z^2]^2}{\mathbb{E}[Z^4]} \geq \left(\frac{3}{4}\right)^2 \cdot \frac{1}{3} = \frac{3}{16}\end{aligned}$$

$$\delta = \frac{1}{4}$$

$$X = Z^2$$

Paley–Zygmund inequality: $\Pr[X \geq \delta \mathbb{E}[X]] \geq (1 - \delta)^2 \frac{\mathbb{E}[X]^2}{\mathbb{E}[X^2]}$, for $X \geq 0$, $\text{Var}[X] < \infty$, $\delta \in (0, 1)$

The power method

The matrix $\mathbf{M}^k$ has:

- Eigenvalues: $0 \leq \lambda_1^k \leq \lambda_2^k \leq \dots \leq \lambda_n^k$
- The same eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

Suppose $\lambda_{i^*} \leq (1 - \epsilon) \cdot \lambda_n < \lambda_{i^*+1}$.

$$\mathbf{x}_0 = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_i \mathbf{v}_i + c_{i+1} \mathbf{v}_{i+1} + \dots + \mathbf{c}_n \mathbf{v}_n$$



Assume $|c_n| = |\langle \mathbf{x}_0, \mathbf{v}_n \rangle| \geq \frac{1}{2}$.

$$\mathbf{x} = \mathbf{M}^k \mathbf{x}_0$$

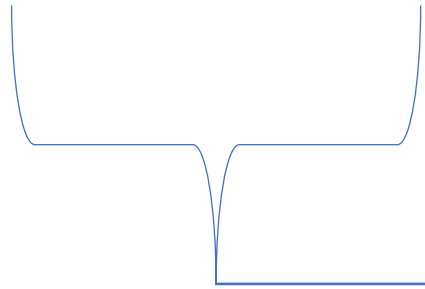
$$\begin{aligned} R_{\mathbf{M}}(\mathbf{x}) &= \frac{\mathbf{x}^\top \mathbf{M} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} \\ &= \frac{\mathbf{x}_0^\top \mathbf{M}^{2k+1} \mathbf{x}_0}{\mathbf{x}_0^\top \mathbf{M}^{2k} \mathbf{x}_0} \\ &= \frac{\sum_{i=1}^n c_i^2 \lambda_i^{2k+1}}{\sum_{i=1}^n c_i^2 \lambda_i^{2k}} \end{aligned}$$

The power method

The matrix $\mathbf{M}^k$ has:

- Eigenvalues: $0 \leq \lambda_1^k \leq \lambda_2^k \leq \dots \leq \lambda_n^k$
- The same eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

Suppose $\lambda_{i^*} \leq (1 - \epsilon) \cdot \lambda_n < \lambda_{i^*+1}$.



$$\mathbf{x}_0 = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_i \mathbf{v}_i + c_{i+1} \mathbf{v}_{i+1} + \dots + \mathbf{c}_n \mathbf{v}_n$$



Assume $|c_n| = |\langle \mathbf{x}_0, \mathbf{v}_n \rangle| \geq \frac{1}{2}$.

$$\begin{aligned} R_{\mathbf{M}}(\mathbf{x}) &= \frac{\mathbf{x}^\top \mathbf{M} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} \\ &= \frac{\mathbf{x}_0^\top \mathbf{M}^{2k+1} \mathbf{x}_0}{\mathbf{x}_0^\top \mathbf{M}^{2k} \mathbf{x}_0} \\ &= \frac{\sum_{i=1}^n c_i^2 \lambda_i^{2k+1}}{\sum_{i=1}^n c_i^2 \lambda_i^{2k}} \\ &\geq \frac{(1 - \epsilon) \cdot \lambda_n \cdot \sum_{i=i^*+1}^n c_i^2 \lambda_i^{2k}}{\sum_{i=1}^{i^*} c_i^2 \lambda_i^{2k} + \sum_{i=i^*+1}^n c_i^2 \lambda_i^{2k}} \end{aligned}$$

The power method

$$\mathbf{x}_0 = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_i \mathbf{v}_i + c_{i+1} \mathbf{v}_{i+1} + \cdots + c_n \mathbf{v}_n$$

Assume $|c_n| = |\langle \mathbf{x}_0, \mathbf{v}_n \rangle| \geq \frac{1}{2}$.

The matrix $\mathbf{M}^k$ has:

- Eigenvalues: $0 \leq \lambda_1^k \leq \lambda_2^k \leq \cdots \leq \lambda_n^k$
- The same eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

Suppose $\lambda_{i^*} \leq (1 - \epsilon) \cdot \lambda_n < \lambda_{i^*+1}$.

$$\begin{aligned} \sum_{i=1}^n c_i^2 &= \|\mathbf{x}_0\|^2 = n \\ 4 \cdot c_n^2 &\geq 1 \\ \sum_{i=1}^{i^*} c_i^2 \lambda_i^{2k} &\leq (1 - \epsilon)^{2k} \cdot \lambda_n^{2k} \cdot \sum_{i=1}^n c_i^2 \\ &= (1 - \epsilon)^{2k} \cdot n \cdot \lambda_n^{2k} \\ &\leq (1 - \epsilon)^{2k} \cdot 4n \cdot c_n^2 \lambda_n^{2k} \\ &\leq (1 - \epsilon)^{2k} \cdot 4n \cdot \sum_{i=i^*+1}^n c_i^2 \lambda_i^{2k} \end{aligned}$$

$$\begin{aligned} R_M(\mathbf{x}) &= \frac{\mathbf{x}^\top \mathbf{M} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} \\ &= \frac{\mathbf{x}_0^\top \mathbf{M}^{2k+1} \mathbf{x}_0}{\mathbf{x}_0^\top \mathbf{M}^{2k} \mathbf{x}_0} \\ &= \frac{\sum_{i=1}^n c_i^2 \lambda_i^{2k+1}}{\sum_{i=1}^n c_i^2 \lambda_i^{2k}} \\ &\geq \frac{(1 - \epsilon) \cdot \lambda_n \cdot \sum_{i=i^*+1}^n c_i^2 \lambda_i^{2k}}{\sum_{i=1}^{i^*} c_i^2 \lambda_i^{2k} + \sum_{i=i^*+1}^n c_i^2 \lambda_i^{2k}} \\ &\geq \frac{(1 - \epsilon) \cdot \lambda_n \cdot \sum_{i=i^*+1}^n c_i^2 \lambda_i^{2k}}{(1 + (1 - \epsilon)^{2k} \cdot 4n) \sum_{i=i^*+1}^n c_i^2 \lambda_i^{2k}} \end{aligned}$$

The power method

The matrix $\mathbf{M}^k$ has:

- Eigenvalues: $0 \leq \lambda_1^k \leq \lambda_2^k \leq \dots \leq \lambda_n^k$
- The same eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$

Suppose $\lambda_{i^*} \leq (1 - \epsilon) \cdot \lambda_n < \lambda_{i^*+1}$.

$$\mathbf{x}_0 = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_i \mathbf{v}_i + c_{i+1} \mathbf{v}_{i+1} + \dots + c_n \mathbf{v}_n$$



Assume $|c_n| = |\langle \mathbf{x}_0, \mathbf{v}_n \rangle| \geq \frac{1}{2}$.

$$\begin{aligned} R_{\mathbf{M}}(\mathbf{x}) &= \frac{\mathbf{x}^\top \mathbf{M} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} \\ &= \frac{\mathbf{x}_0^\top \mathbf{M}^{2k+1} \mathbf{x}_0}{\mathbf{x}_0^\top \mathbf{M}^{2k} \mathbf{x}_0} \\ &= \frac{\sum_{i=1}^n c_i^2 \lambda_i^{2k+1}}{\sum_{i=1}^n c_i^2 \lambda_i^{2k}} \\ &\geq \frac{(1 - \epsilon) \cdot \lambda_n \cdot \sum_{i=i^*+1}^n c_i^2 \lambda_i^{2k}}{\sum_{i=1}^{i^*} c_i^2 \lambda_i^{2k} + \sum_{i=i^*+1}^n c_i^2 \lambda_i^{2k}} \\ &\geq \frac{(1 - \epsilon) \cdot \lambda_n \cdot \sum_{i=i^*+1}^n c_i^2 \lambda_i^{2k}}{(1 + (1 - \epsilon)^{2k} \cdot 4n) \sum_{i=i^*+1}^n c_i^2 \lambda_i^{2k}} \\ &= \frac{(1 - \epsilon) \cdot \lambda_n}{1 + (1 - \epsilon)^{2k} \cdot 4n} \\ &\in \left(1 - O(\epsilon)\right) \cdot \lambda_n \end{aligned}$$

$$k \in O\left(\frac{\log n}{\epsilon}\right)$$

The power method

$$\mathbf{x}_0 = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_i \mathbf{v}_i + c_{i+1} \mathbf{v}_{i+1} + \cdots + c_n \mathbf{v}_n$$



Assume $|c_n| = |\langle \mathbf{x}_0, \mathbf{v}_n \rangle| \geq \frac{1}{2}$.

This happens with probability $\geq \frac{3}{16}$.

Claim: There is a choice of parameter $k \in O\left(\frac{\log n}{\epsilon}\right)$ such that $\Pr[R_M(\mathbf{x}) \geq (1 - \epsilon) \cdot \lambda_n] \geq \frac{3}{16}$.

with a change of variable: $\epsilon' \in \Theta(\epsilon)$

$$\begin{aligned} R_M(\mathbf{x}) &= \frac{\mathbf{x}^\top \mathbf{M} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} \\ &= \frac{\mathbf{x}_0^\top \mathbf{M}^{2k+1} \mathbf{x}_0}{\mathbf{x}_0^\top \mathbf{M}^{2k} \mathbf{x}_0} \\ &= \frac{\sum_{i=1}^n c_i^2 \lambda_i^{2k+1}}{\sum_{i=1}^n c_i^2 \lambda_i^{2k}} \\ &\geq \frac{(1 - \epsilon) \cdot \lambda_n \cdot \sum_{i=i^*+1}^n c_i^2 \lambda_i^{2k}}{\sum_{i=1}^{i^*} c_i^2 \lambda_i^{2k} + \sum_{i=i^*+1}^n c_i^2 \lambda_i^{2k}} \\ &\geq \frac{(1 - \epsilon) \cdot \lambda_n \cdot \sum_{i=i^*+1}^n c_i^2 \lambda_i^{2k}}{(1 + (1 - \epsilon)^{2k} \cdot 4n) \sum_{i=i^*+1}^n c_i^2 \lambda_i^{2k}} \\ &= \frac{(1 - \epsilon) \cdot \lambda_n}{1 + (1 - \epsilon)^{2k} \cdot 4n} \\ &\in (1 - O(\epsilon)) \cdot \lambda_n \end{aligned}$$

Second largest eigenvector

- Let $\mathbf{M}$ be a real symmetric $(n \times n)$ -matrix.
 - Eigenvalues: $0 \leq \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$
 - Eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ (orthonormal vectors)
- **Done:** Find a vector $\mathbf{x}$ with $R_{\mathbf{M}}(\mathbf{x}) \geq (1 - \epsilon) \cdot \lambda_n$.
- **Next:** The second largest eigenvector $\mathbf{v}_{n-1}$.

Given $\mathbf{v}_n$, the goal is to find a vector $\mathbf{x} \perp \mathbf{v}_n$ with $R_{\mathbf{M}}(\mathbf{x}) \geq (1 - \epsilon) \cdot \lambda_{n-1}$.

Second largest eigenvector

Claim: for any unit vector $\mathbf{v}$, $\Pr \left[|\langle \mathbf{x}_0, \mathbf{v} \rangle| \geq \frac{1}{2} \right] \geq \frac{3}{16}$

$$\langle \mathbf{x}'_0, \mathbf{v}_{n-1} \rangle = \langle \mathbf{x}_0, \mathbf{v}_{n-1} \rangle$$

We still have:

$$\Pr \left[|\langle \mathbf{x}'_0, \mathbf{v}_{n-1} \rangle| \geq \frac{1}{2} \right] \geq \frac{3}{16}$$

The same analysis:

There is a choice of parameter $k \in O\left(\frac{\log n}{\epsilon}\right)$ such that $\Pr[R_M(\mathbf{x}) \geq (1 - \epsilon) \cdot \lambda_{n-1}] \geq \frac{3}{16}$.

Algorithm POWER:

Input: a matrix $\mathbf{M}$ and a parameter k .

- Pick $\mathbf{x}_0 \in \{-1, 1\}^n$ uniformly at random.
- Return $\mathbf{x} = \mathbf{M}^k \mathbf{x}_0$.

- Pick $\mathbf{x}_0 \in \{-1, 1\}^n$ uniformly at random.
- $\mathbf{x}'_0 = \mathbf{x}_0 - \langle \mathbf{x}_0, \mathbf{v}_n \rangle \cdot \mathbf{v}_n$
- Return $\mathbf{x} = \mathbf{M}^k \mathbf{x}'_0$.

Approximating $\mathbf{v}_1$ and $\mathbf{v}_2$

- Let $\mathbf{M}$ be a real symmetric $(n \times n)$ -matrix.
 - Eigenvalues: $0 \leq \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$
 - Eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ (orthonormal vectors)
- Suppose we know an upper bound $c \geq \lambda_n$
- Consider $c\mathbf{I} - \mathbf{M}$:
 - Same eigenvectors: $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$
 - Eigenvalues: $c - \lambda_1 \geq c - \lambda_2 \geq \dots \geq c - \lambda_n \geq 0$

Now we can use the same method to approximate $\mathbf{v}_1$ and $\mathbf{v}_2$.

Computing a low-conductance cut

- Approximating the second largest eigenvector of $2\mathbf{I} - \mathbf{N}$:
 - We get a vector $\mathbf{x} \perp \mathbf{1}$ with

$$R_{2\mathbf{I}-\mathbf{N}}(\mathbf{x}) = 2 - R_{\mathbf{N}}(\mathbf{x}) \geq (1 - \epsilon) \cdot (2 - \lambda_2)$$



$$R_{\mathbf{N}}(\mathbf{x}) \leq 2 - (1 - \epsilon) \cdot (2 - \lambda_2) \leq \lambda_2 + 2\epsilon$$



$$\epsilon' = 2\epsilon$$

We get a cut with conductance at most $\sqrt{2(\lambda_2 + \epsilon')}$.

Random walks on graphs

- Given an undirected unweighted connected graph G and a vertex u , the result of one step of a random walk from u is a uniformly random neighbor of u in G .
- Random walk matrix: $\mathbf{W} = \mathbf{D}^{-1}\mathbf{A}$
 - $\mathbf{W}[u, v] = \begin{cases} \frac{1}{\deg(u)} & \{u, v\} \in E \\ 0 & \{u, v\} \notin E \end{cases}$

$\mathbf{x}$ is the current probability distribution over the vertices.

$\mathbf{W}^T \mathbf{x}$ is the probability distribution resulting from one step of a random walk.

Stationary distribution

- A probability distribution x over the vertices is **stationary** if $W^T x = x$.

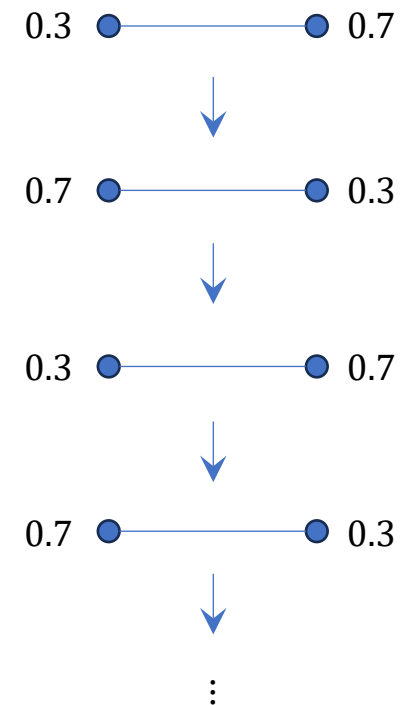
- Does a stationary distribution exist?

Yes: $x(u) = \frac{\deg(u)}{2|E|}$

- Do random walks converge to a stationary distribution?

No for bipartite graphs.

How to fix it?



Lazy random walks

- In one step of a **lazy** random walk:
 - With probability $\frac{1}{2}$, you stay at your current place.
 - With probability $\frac{1}{2}$, you perform a standard random walk step.

- Lazy random walk matrix: $\widetilde{\mathbf{W}} = \frac{1}{2} (\mathbf{I} + \mathbf{D}^{-1} \mathbf{A})$

$$\bullet \quad \widetilde{\mathbf{W}}[u, v] = \begin{cases} \frac{1}{2 \deg(u)} & \{u, v\} \in E \\ \frac{1}{2} & u = v \\ 0 & \text{otherwise} \end{cases}$$

$x(u) = \frac{\deg(u)}{2|E|}$ is still a stationary distribution: $\widetilde{\mathbf{W}}^T \mathbf{x} = \mathbf{x}$

What about convergence?

Regular graphs

- For simplicity, we restrict our discussion to d -regular graphs.

Lazy random walk matrix: $\widetilde{\mathbf{W}} = \frac{1}{2}(\mathbf{I} + \mathbf{D}^{-1}\mathbf{A}) = \frac{1}{2}\mathbf{I} + \frac{1}{2d}\mathbf{A} = \mathbf{I} - \frac{1}{2}\mathbf{N}$

$$\mathbf{N} = \mathbf{I} - \frac{1}{d}\mathbf{A}$$

$$\widetilde{\mathbf{W}}[u, v] = \begin{cases} \frac{1}{2 \deg(u)} = \frac{1}{2d} & \{u, v\} \in E \\ \frac{1}{2} & u = v \\ 0 & \text{otherwise} \end{cases}$$

$x(u) = \frac{\deg(u)}{2|E|} = \frac{1}{n}$ is a stationary distribution : $\widetilde{\mathbf{W}}^\top \mathbf{x} = \mathbf{x}$

Uniform distribution: $\mathbf{x} = \frac{1}{n}\mathbf{1}$

$$\widetilde{\mathbf{W}} = \widetilde{\mathbf{W}}^\top$$

Convergence

Lazy random walk on a regular graph
converges to the uniform distribution.

Claim: $\forall \mathbf{x} \in \mathbb{R}^n$ with $\sum_{i=1}^n x_i = 1$, we have: $\lim_{t \rightarrow \infty} \left(I - \frac{1}{2} N \right)^t \mathbf{x} = \frac{1}{n} \mathbf{1}$

Eigenvalue range for a connected d -regular graphs:

- $N = I - \frac{1}{d} A : [0, 2]$
 - $\mathbf{v}_1 = \frac{1}{\sqrt{n}} \mathbf{1}$ is an eigenvector with eigenvalue $\lambda_1 = 0$.
 - $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_n$.
- $I - \frac{1}{2} N : [0, 1]$
 - $\mathbf{v}_1 = \frac{1}{\sqrt{n}} \mathbf{1}$ is an eigenvector with eigenvalue $1 - \frac{\lambda_1}{2} = 1$.
 - All other eigenvalues are within the range $\left[0, 1 - \frac{\lambda_2}{2}\right]$.

$$\begin{aligned} & \left(I - \frac{1}{2} N \right)^t \mathbf{x} \\ &= \sum_{i=1}^n \left(1 - \frac{\lambda_i}{2} \right)^t \langle \mathbf{x}, \mathbf{v}_i \rangle \mathbf{v}_i \\ &= \underbrace{\langle \mathbf{x}, \mathbf{v}_1 \rangle \mathbf{v}_1}_{1 - \frac{\lambda_1}{2} = 1} + \sum_{i=2}^n \left(1 - \frac{\lambda_i}{2} \right)^t \langle \mathbf{x}, \mathbf{v}_i \rangle \mathbf{v}_i \end{aligned}$$

Convergence

Lazy random walk on a regular graph
converges to the uniform distribution.

Claim: $\forall \mathbf{x} \in \mathbb{R}^n$ with $\sum_{i=1}^n x_i = 1$, we have: $\lim_{t \rightarrow \infty} \left(I - \frac{1}{2} N \right)^t \mathbf{x} = \frac{1}{n} \mathbf{1}$

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 - All other eigenvalues are within the range $\left[0, 1 - \frac{\lambda_2}{2}\right]$.

$$\begin{aligned} & \left(I - \frac{1}{2} N \right)^t \mathbf{x} \\ &= \sum_{i=1}^n \left(1 - \frac{\lambda_i}{2} \right)^t \langle \mathbf{x}, \mathbf{v}_i \rangle \mathbf{v}_i \\ &= \langle \mathbf{x}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \sum_{i=2}^n \left(1 - \frac{\lambda_i}{2} \right)^t \langle \mathbf{x}, \mathbf{v}_i \rangle \mathbf{v}_i \\ &= \frac{1}{n} \mathbf{1} + \underbrace{\sum_{i=2}^n \left(1 - \frac{\lambda_i}{2} \right)^t \langle \mathbf{x}, \mathbf{v}_i \rangle \mathbf{v}_i}_{\rightarrow 0 \text{ as } t \rightarrow \infty} \end{aligned}$$

$\sum_{i=1}^n x_i = 1$ $\rightarrow 0 \text{ as } t \rightarrow \infty$

Convergence rate

- How fast does a lazy random walk converge to the uniform distribution?
- A standard way to measure the distance between two distributions:

Total variation distance:

$$d_{\text{TV}}(\mathbf{x}, \mathbf{y}) = \frac{1}{2} \sum_v |x_v - y_v| = \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|_1$$

The **mixing time** $\tau_{\text{mix}}(G, \epsilon)$ is the smallest integer t such that:

- Starting from any initial distribution $\mathbf{x}$ over the vertex set, after t steps of lazy random walk, the total variation distance to the uniform distribution is at most ϵ .

Convergence rate

claim: $\tau_{\text{mix}}(G, \epsilon) \in O\left(\frac{\log \frac{n}{\epsilon}}{\lambda_2}\right)$

- Initial distribution: x
- After t steps of lazy random walk: $\left(I - \frac{1}{2}N\right)^t x$
- **Goal:** $d_{\text{TV}}\left(\frac{1}{n}\mathbf{1}, \left(I - \frac{1}{2}N\right)^t x\right) \leq \epsilon$.

$$\frac{1}{2} \left\| \frac{1}{n}\mathbf{1} - \left(I - \frac{1}{2}N\right)^t x \right\|_1 = \frac{1}{2} \left\| \sum_{i=2}^n \left(1 - \frac{\lambda_i}{2}\right)^t \langle x, v_i \rangle v_i \right\|_1$$

Recall: $\left(I - \frac{1}{2}N\right)^t x = \frac{1}{n}\mathbf{1} + \sum_{i=2}^n \left(1 - \frac{\lambda_i}{2}\right)^t \langle x, v_i \rangle v_i$

Convergence rate

claim: $\tau_{\text{mix}}(G, \epsilon) \in O\left(\frac{\log \frac{n}{\epsilon}}{\lambda_2}\right)$

- Initial distribution: x
- After t steps of lazy random walk: $\left(I - \frac{1}{2}N\right)^t x$
- **Goal:** $d_{\text{TV}}\left(\frac{1}{n}\mathbf{1}, \left(I - \frac{1}{2}N\right)^t x\right) \leq \epsilon$.

$$\frac{1}{2} \left\| \frac{1}{n}\mathbf{1} - \left(I - \frac{1}{2}N\right)^t x \right\|_1 = \frac{1}{2} \left\| \sum_{i=2}^n \left(1 - \frac{\lambda_i}{2}\right)^t \langle x, v_i \rangle v_i \right\|_1 \leq \frac{\sqrt{n}}{2} \left\| \sum_{i=2}^n \left(1 - \frac{\lambda_i}{2}\right)^t \langle x, v_i \rangle v_i \right\|_2$$

Relation between 1-norm and 2-norm for vectors in $\mathbb{R}^n$:

$$\|v\|_2 \leq \|v\|_1 \leq \sqrt{n}\|v\|_2$$

Convergence rate

claim: $\tau_{\text{mix}}(G, \epsilon) \in O\left(\frac{\log \frac{n}{\epsilon}}{\lambda_2}\right)$

- Initial distribution: $\mathbf{x}$
- After t steps of lazy random walk: $\left(I - \frac{1}{2}N\right)^t \mathbf{x}$
- **Goal:** $d_{\text{TV}}\left(\frac{1}{n}\mathbf{1}, \left(I - \frac{1}{2}N\right)^t \mathbf{x}\right) \leq \epsilon$.

$$\begin{aligned} \frac{1}{2} \left\| \frac{1}{n}\mathbf{1} - \left(I - \frac{1}{2}N\right)^t \mathbf{x} \right\|_1 &= \frac{1}{2} \left\| \sum_{i=2}^n \left(1 - \frac{\lambda_i}{2}\right)^t \langle \mathbf{x}, \mathbf{v}_i \rangle \mathbf{v}_i \right\|_1 \leq \frac{\sqrt{n}}{2} \left\| \sum_{i=2}^n \left(1 - \frac{\lambda_i}{2}\right)^t \langle \mathbf{x}, \mathbf{v}_i \rangle \mathbf{v}_i \right\|_2 \\ &= \frac{\sqrt{n}}{2} \cdot \sqrt{\sum_{i=2}^n \left(1 - \frac{\lambda_i}{2}\right)^{2t} \langle \mathbf{x}, \mathbf{v}_i \rangle^2} \leq \frac{\sqrt{n}}{2} \cdot \left(1 - \frac{\lambda_2}{2}\right)^t \sqrt{\sum_{i=2}^n \langle \mathbf{x}, \mathbf{v}_i \rangle^2} \leq \frac{\sqrt{n}}{2} \cdot \left(1 - \frac{\lambda_2}{2}\right)^t \|\mathbf{x}\|_2 \end{aligned}$$

$\forall i \geq 2, \quad 1 - \frac{\lambda_i}{2} \leq 1 - \frac{\lambda_2}{2}$

$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n \langle \mathbf{x}, \mathbf{v}_i \rangle^2}$

Convergence rate

claim: $\tau_{\text{mix}}(G, \epsilon) \in O\left(\frac{\log \frac{n}{\epsilon}}{\lambda_2}\right)$

- Initial distribution: $\mathbf{x}$
- After t steps of lazy random walk: $\left(I - \frac{1}{2}\mathbf{N}\right)^t \mathbf{x}$
- **Goal:** $d_{\text{TV}}\left(\frac{1}{n}\mathbf{1}, \left(I - \frac{1}{2}\mathbf{N}\right)^t \mathbf{x}\right) \leq \epsilon$.

$$\begin{aligned} \frac{1}{2} \left\| \frac{1}{n}\mathbf{1} - \left(I - \frac{1}{2}\mathbf{N}\right)^t \mathbf{x} \right\|_1 &= \frac{1}{2} \left\| \sum_{i=2}^n \left(1 - \frac{\lambda_i}{2}\right)^t \langle \mathbf{x}, \mathbf{v}_i \rangle \mathbf{v}_i \right\|_1 \leq \frac{\sqrt{n}}{2} \left\| \sum_{i=2}^n \left(1 - \frac{\lambda_i}{2}\right)^t \langle \mathbf{x}, \mathbf{v}_i \rangle \mathbf{v}_i \right\|_2 \\ &= \frac{\sqrt{n}}{2} \cdot \sqrt{\sum_{i=2}^n \left(1 - \frac{\lambda_i}{2}\right)^{2t} \langle \mathbf{x}, \mathbf{v}_i \rangle^2} \leq \frac{\sqrt{n}}{2} \cdot \left(1 - \frac{\lambda_2}{2}\right)^t \sqrt{\sum_{i=2}^n \langle \mathbf{x}, \mathbf{v}_i \rangle^2} \leq \frac{\sqrt{n}}{2} \cdot \left(1 - \frac{\lambda_2}{2}\right)^t \|\mathbf{x}\|_2 \leq \frac{\sqrt{n}}{2} \cdot \left(1 - \frac{\lambda_2}{2}\right)^t \end{aligned}$$

$\|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1 = 1$

Convergence rate

$$\text{claim: } \tau_{\text{mix}}(G, \epsilon) \in O\left(\frac{\log \frac{n}{\epsilon}}{\lambda_2}\right) \subseteq O\left(\frac{\log \frac{n}{\epsilon}}{\Phi(G)^2}\right)$$

From now on, we return to **general graphs**, where the claim still holds, with the uniform distribution replaced by the degree distribution: $x(u) = \frac{\deg(u)}{2|E|}$.

$$\text{Exercise: } \tau_{\text{mix}}\left(G, \frac{1}{4}\right) \in \Omega\left(\frac{1}{\Phi(G)}\right)$$

Proof sketch:

- Consider a cut with conductance $\Phi(G)$.
- Start a lazy random walk at a random vertex in the smaller side of the cut.
- In t steps, the amount of probability mass transferred to the other side is at most $t \cdot \Phi(G)$.

Cover time

- How long does it take for a random walk to visit all vertices?
 - Set $\epsilon = \frac{1}{8|E|}$.
 - Doing a random walk for $\tau_{\text{mix}}(G, \epsilon)$ steps. $\in O\left(\frac{\log n}{\Phi(G)^2}\right) \subseteq O(|E|^2 \log n)$
 - The probability that we are at vertex v is at least $\frac{\deg(v)}{2|E|} - 2\epsilon \geq \frac{1}{4|E|}$.

Repeating this for $4|E| \cdot 2 \ln n$ times

$O(|E|^3 \log^2 n)$ steps are enough.

Vertex v is visited with probability $\geq 1 - \left(1 - \frac{1}{4|E|}\right)^{4|E| \cdot 2 \ln n} \geq 1 - n^{-2}$

Every vertex is visited with probability $\geq 1 - n^{-1}$

Application 1: Space-efficient algorithm

- Consider this problem:
 - Input: An n -vertex m -edge graph $G = (V, E)$ and two vertices s and t .
 - Goal: Decide whether s and t are connected.

- **Easy:** Solvable in linear time by a BFS from s .

Maintaining a search tree requires $O(n)$ words of $O(\log n)$ bits.

- **Difficult:** Is it possible to solve the problem in polynomial time with a space complexity that is significantly smaller than $O(n)$?

Yes: Just do a random walk from s for $O(|E|^3 \log^2 n)$ steps and see if t is reached.

- Storing the state of a random walk costs only one word of $O(\log n)$ bits.

Application 2: Sampling

- **Easy:** Computing a spanning tree can be done in polynomial time.
- **Difficult:** Sampling a spanning tree nearly uniformly at random.

Given a spanning tree T , Consider the following **flip** operation:

- Choose a non-tree-edge e .
- Consider the unique cycle C in $T + e$.
- Choose a tree-edge $e' \in C$.
- Return the spanning tree $T' = T + e - e'$.

Consider a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ over the set of all spanning trees $\mathcal{V}$:

- $\{T, T'\} \in \mathcal{E}$ if T' can be reached from T by a flip operation.

While the size of $\mathcal{G}$ is exponential in n , it is a very good expander, with its mixing time polynomial in n .

To sample a spanning tree nearly uniformly at random, it suffices to simulate a random walk in $\mathcal{G}$ for a polynomial number of steps, and this can be done in polynomial time.

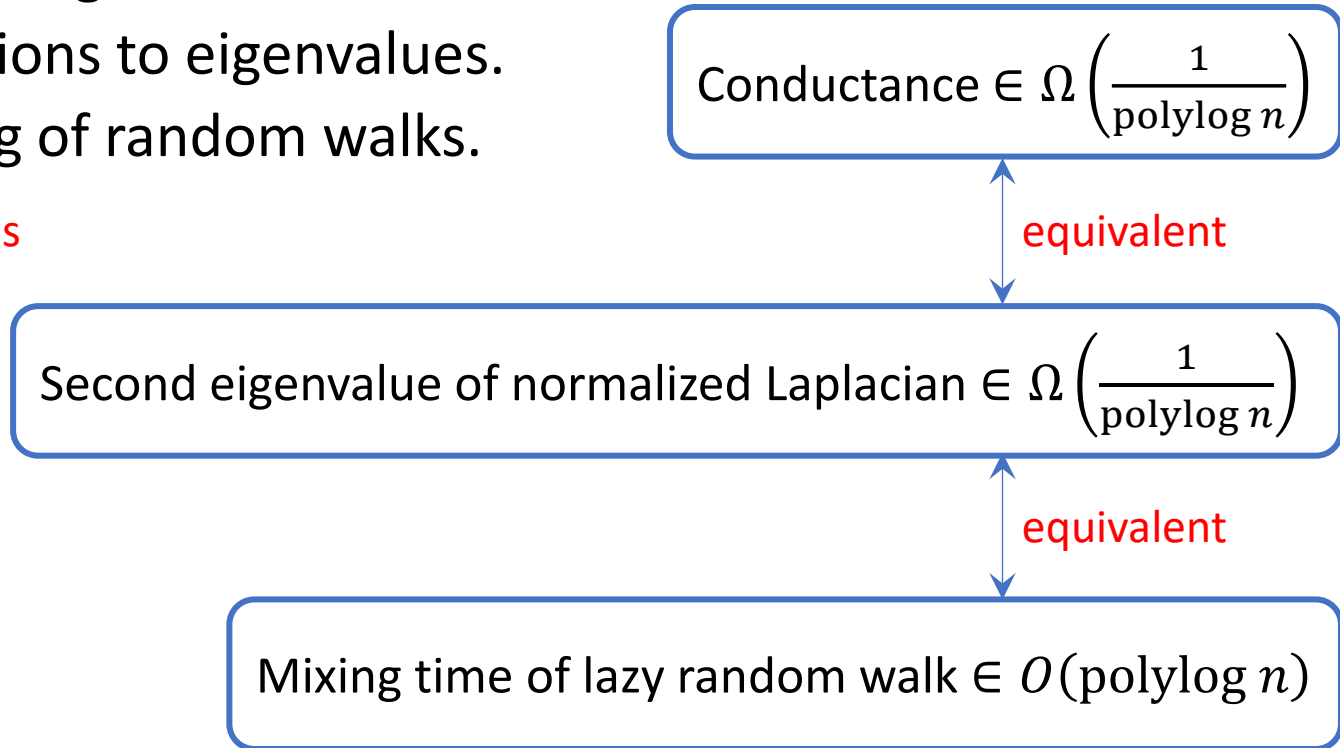
<https://arxiv.org/pdf/2004.07220>

Outlook

- We have discussed several key aspects of expanders:

- **Connectivity:** robustness against deletions.
- **Linear algebra:** connections to eigenvalues.
- **Probability:** rapid mixing of random walks.

with many tools and applications



Quasirandomness

- **Observation:**

- For n -vertex d -regular graphs, the following two statements are equivalent.

For any cut $(S, V \setminus S)$, we have:
 $|E(S, V \setminus S)| \in \Theta\left(\frac{d}{n} \cdot |S| |V \setminus S|\right)$

$G = (V, E)$ is an $\Omega(1)$ -expander.

This is the expected number of edges in $E(S, V \setminus S)$ if G is a random n -vertex d -regular graph.

This can be seen as a **quasirandomness** property.

Question: Can we extend this to $E(S, T)$ for any two subsets $S \subseteq V$ and $T \subseteq V$?

Notation

- For simplicity, we restrict ourselves to n -vertex d -regular graphs.

$$e(A, B) = |\{(a, b) \in A \times B : \{a, b\} \in E\}|$$

- This is the number of edges between A and B .
- If $\{u, v\} \in E$ satisfies $\{u, v\} \subseteq A \cap B$, then $\{u, v\}$ is counted twice.

$$\sigma_2 = \max\{|\lambda_2|, |\lambda_3|, \dots, |\lambda_n|\},$$

- where $d = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq -d$ are the eigenvalues of the adjacency matrix A .

Expander mixing lemma

- For simplicity, we restrict ourselves to n -vertex d -regular graphs.

$$e(A, B) = |\{(a, b) \in A \times B : \{a, b\} \in E\}|$$

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- where $d = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq -d$ are the eigenvalues of the adjacency matrix A .

Expander mixing lemma:

$$\left| e(S, T) - \frac{d}{n} \cdot |S| |T| \right| \leq \sigma_2 \cdot \sqrt{|S| |T|}$$

The number of edges between S and T is close to the expected number of edges between them in a random d -regular graph.

Proof of the lemma

- $e(S, T) = \mathbf{1}_S^\top \mathbf{A} \mathbf{1}_T$
- $|S||T| = \mathbf{1}_S^\top \mathbf{J} \mathbf{1}_T$, where $\mathbf{J}$ is the all-1 ($n \times n$) matrix.

$$\left| e(S, T) - \frac{d}{n} \cdot |S||T| \right| = \left| \mathbf{1}_S^\top \mathbf{A} \mathbf{1}_T - \frac{d}{n} \cdot \mathbf{1}_S^\top \mathbf{J} \mathbf{1}_T \right| = \left| \mathbf{1}_S^\top \left(\mathbf{A} - \frac{d}{n} \mathbf{J} \right) \mathbf{1}_T \right|$$

Expander mixing lemma:

$$\left| e(S, T) - \frac{d}{n} \cdot |S||T| \right| \leq \sigma_2 \cdot \sqrt{|S||T|}$$

Proof of the lemma

- $e(S, T) = \mathbf{1}_S^\top \mathbf{A} \mathbf{1}_T$
- $|S||T| = \mathbf{1}_S^\top \mathbf{J} \mathbf{1}_T$, where $\mathbf{J}$ is the all-1 ($n \times n$) matrix.

$$\begin{aligned} \left| e(S, T) - \frac{d}{n} \cdot |S||T| \right| &= \left| \mathbf{1}_S^\top \mathbf{A} \mathbf{1}_T - \frac{d}{n} \cdot \mathbf{1}_S^\top \mathbf{J} \mathbf{1}_T \right| = \left| \mathbf{1}_S^\top \left(\mathbf{A} - \frac{d}{n} \mathbf{J} \right) \mathbf{1}_T \right| \\ &\leq \|\mathbf{1}_S\| \cdot \left\| \left(\mathbf{A} - \frac{d}{n} \mathbf{J} \right) \mathbf{1}_T \right\| \leq \|\mathbf{1}_S\| \cdot \sigma_2 \cdot \|\mathbf{1}_T\| \end{aligned}$$

$$\begin{aligned} \left\| \left(\mathbf{A} - \frac{d}{n} \mathbf{J} \right) \mathbf{1}_T \right\| &= \left\| \sum_{i=2}^n \lambda_i \langle \mathbf{1}_T, \mathbf{v}_i \rangle \mathbf{v}_i \right\| \\ &= \sqrt{\sum_{i=2}^n \lambda_i^2 \langle \mathbf{1}_T, \mathbf{v}_i \rangle^2} \leq \sigma_2 \cdot \sqrt{\sum_{i=2}^n \langle \mathbf{1}_T, \mathbf{v}_i \rangle^2} \leq \sigma_2 \cdot \|\mathbf{1}_T\| \end{aligned}$$

$$\sigma_2 = \max\{|\lambda_2|, |\lambda_3|, \dots, |\lambda_n|\}$$

Expander mixing lemma:

$$\left| e(S, T) - \frac{d}{n} \cdot |S||T| \right| \leq \sigma_2 \cdot \sqrt{|S||T|}$$

Eigenvalues

Eigenvectors	$\mathbf{A}$	$\frac{d}{n} \mathbf{J}$	$\mathbf{A} - \frac{d}{n} \mathbf{J}$
$\mathbf{v}_1 = \frac{1}{\sqrt{n}} \mathbf{1}$	$\lambda_1 = d$	d	0
$\mathbf{v}_2$	λ_2	0	λ_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\mathbf{v}_n$	λ_n	0	λ_n

Proof of the lemma

- $e(S, T) = \mathbf{1}_S^\top \mathbf{A} \mathbf{1}_T$
- $|S||T| = \mathbf{1}_S^\top \mathbf{J} \mathbf{1}_T$, where $\mathbf{J}$ is the all-1 ($n \times n$) matrix.

$$\begin{aligned} \left| e(S, T) - \frac{d}{n} \cdot |S||T| \right| &= \left| \mathbf{1}_S^\top \mathbf{A} \mathbf{1}_T - \frac{d}{n} \cdot \mathbf{1}_S^\top \mathbf{J} \mathbf{1}_T \right| = \left| \mathbf{1}_S^\top \left(\mathbf{A} - \frac{d}{n} \mathbf{J} \right) \mathbf{1}_T \right| \\ &\leq \|\mathbf{1}_S\| \cdot \left\| \left(\mathbf{A} - \frac{d}{n} \mathbf{J} \right) \mathbf{1}_T \right\| \leq \|\mathbf{1}_S\| \cdot \sigma_2 \cdot \|\mathbf{1}_T\| = \sigma_2 \cdot \sqrt{|S||T|} \end{aligned}$$

Expander mixing lemma:

$$\left| e(S, T) - \frac{d}{n} \cdot |S||T| \right| \leq \sigma_2 \cdot \sqrt{|S||T|}$$

Ramanujan graphs

- How small σ_2 can be?
 - **Answer:** $2\sqrt{d-1}$



Alon-Boppana bound: For every d and $\epsilon > 0$, there exists n_0 such that all graphs with $\geq n_0$ vertices have $\sigma_2 > 2\sqrt{d-1} - \epsilon$.

Expander mixing lemma:

$$\left| e(S, T) - \frac{d}{n} \cdot |S| |T| \right| \leq \sigma_2 \cdot \sqrt{|S| |T|}$$

Proof of a weaker bound:

$\sigma_2 \in \Omega(\sqrt{d})$ when $d \leq 0.99 \cdot n$

$$\text{tr}(\mathbf{A}^2) = nd$$

The trace of a matrix is the sum of its diagonal entries and equals the sum of its eigenvalues.

$$\text{tr}(\mathbf{A}^2) = \sum_{i=1}^n \lambda_i^2 \leq d^2 + (n-1)\sigma_2^2$$

$$\sigma_2 \geq \sqrt{\frac{nd - d^2}{n-1}} \in \Omega(\sqrt{d})$$

Ramanujan graphs

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The name comes from the Ramanujan–Petersson conjecture, which was used in a construction of some of these graphs.

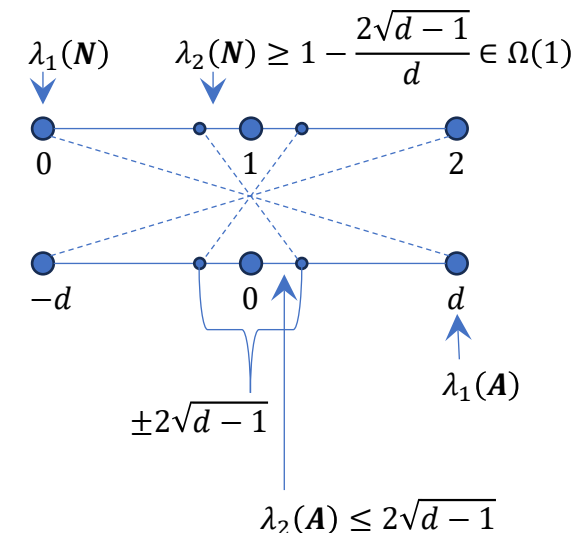
The graphs with $\sigma_2 \leq 2\sqrt{d-1}$ are called **Ramanujan graphs**.

Ramanujan graphs are $\Omega(1)$ -expanders.

Eigenvalues of N :

Eigenvalues of A :

$$N = I - \frac{1}{d}A$$



Ramanujan graphs

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Ramanujan graphs are $\Omega(1)$ -expanders.

Some $\Omega(1)$ -expanders are not Ramanujan.

$\sigma_2 = d$ for any bipartite graph.

Random graphs are nearly Ramanujan

- For every d and $\epsilon > 0$, the probability that a random d -regular graph satisfies $\sigma_2 < 2\sqrt{d-1} + \epsilon$ tends to 1 as $n \rightarrow \infty$.

- Joel Friedman (Duke Mathematical Journal, 2003)

Algebraic constructions

- There is an infinite family of d -regular Ramanujan graphs, whenever $d - 1$ is a prime power.

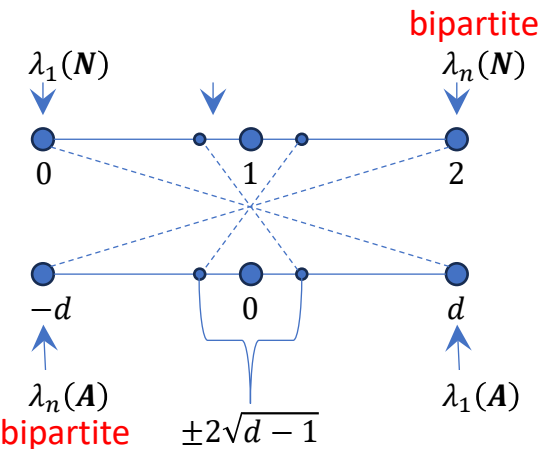
- Alexander Lubotzky, Ralph Phillips, and Peter Sarnak (Combinatorica, 1988)
- Moshe Morgenstern (Journal of Combinatorial Theory, Series B, 1994).

Open problem: Show this for all $d \geq 3$.

Solved for bipartite Ramanujan graphs:

$$\max\{|\lambda_2|, |\lambda_3|, \dots, |\lambda_{n-1}|\} \leq 2\sqrt{d-1}$$

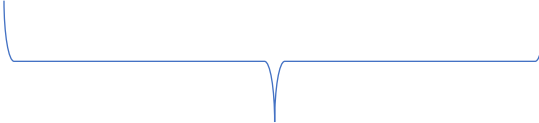
Eigenvalues of N :



- Adam Marcus, Daniel Spielman and Nikhil Srivastava (Annals of Mathematics, 2015)

Properties of Ramanujan graphs

- **Claim:** For any $S, T \subseteq V$, if $|S| \cdot |T| \cdot d \geq 4n^2$, then $e(S, T) > 0$.


$$\sqrt{d|S||T|} \geq 2n$$


$$\frac{d}{n} \cdot |S||T| \geq 2\sqrt{d|S||T|} > \sigma_2 \cdot \sqrt{|S||T|}$$


$$e(S, T) \geq \frac{d}{n} \cdot |S||T| - \sigma_2 \cdot \sqrt{|S||T|} > 0$$

Expander mixing lemma:

$$\left| e(S, T) - \frac{d}{n} \cdot |S||T| \right| \leq \sigma_2 \cdot \sqrt{|S||T|}$$

Ramanujan graphs: $\sigma_2 \leq 2\sqrt{d-1}$

Properties of Ramanujan graphs

- **Claim:** For any $S, T \subseteq V$, if $|S| \cdot |T| \cdot d \geq 4n^2$, then $e(S, T) > 0$.

Exercise: The result is **asymptotically the best possible**.

- If you change $4n^2$ to ϵn^2 for some sufficiently small constant $\epsilon > 0$,
 - then this claim is false for every large enough d -regular graph.

Expander mixing lemma:

$$\left| e(S, T) - \frac{d}{n} \cdot |S| |T| \right| \leq \sigma_2 \cdot \sqrt{|S| |T|}$$

Ramanujan graphs: $\sigma_2 \leq 2\sqrt{d-1}$

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- **Claim:** For any $S, T \subseteq V$, if $|S| \cdot |T| \cdot d \geq 4n^2$, then $e(S, T) > 0$.

Corollary 1: Any independent set S has size $|S| < \frac{2n}{\sqrt{d}}$.

If $|S| \geq \frac{2n}{\sqrt{d}}$, then $|S|^2 d \geq 4n^2$, so $e(S, S) > 0$.

Expander mixing lemma:

$$\left| e(S, T) - \frac{d}{n} \cdot |S| |T| \right| \leq \sigma_2 \cdot \sqrt{|S| |T|}$$

Ramanujan graphs: $\sigma_2 \leq 2\sqrt{d-1}$

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Corollary 2: Chromatic number $> \frac{\sqrt{d}}{2}$.

If a proper coloring with $\leq \frac{\sqrt{d}}{2}$ colors exists, then $n < \frac{\sqrt{d}}{2} \cdot \frac{2n}{\sqrt{d}} = n$, which is impossible.

Expander mixing lemma:

$$\left| e(S, T) - \frac{d}{n} \cdot |S| |T| \right| \leq \sigma_2 \cdot \sqrt{|S| |T|}$$

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Properties of Ramanujan graphs

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Corollary 2: Chromatic number $> \frac{\sqrt{d}}{2}$.

These results have been used to give lower bounds on the number of communication rounds needed to compute certain colorings and independent sets in a distributed network.

Expander mixing lemma:

$$\left| e(S, T) - \frac{d}{n} \cdot |S| |T| \right| \leq \sigma_2 \cdot \sqrt{|S| |T|}$$

- Marijke H.L. Bodlaender, Magnús M. Halldórsson, Christian Konrad, and Fabian Kuhn. “Brief announcement: Local independent set approximation.” *Proceedings of the ACM Symposium on Principles of Distributed Computing* (PODC 2016).
- Nathan Linial. “Locality in distributed graph algorithms.” *SIAM Journal on computing* (1992).

Ramanujan graphs: $\sigma_2 \leq 2\sqrt{d-1}$

Success probability amplification

It uses r random bits and takes t time.

- Consider a randomized algorithm that succeeds with probability $\frac{1}{2}$.
- **Goal:** Amplify the success probability to $1 - f$.
- **Standard solution:** Repeat for $x = \left\lceil \log \frac{1}{f} \right\rceil$ times.  This requires $rx = r \left\lceil \log \frac{1}{f} \right\rceil$ random bits.

Probability of no success is $\frac{1}{2^x} \leq f$.

Question: Is it possible to reduce the failure probability without using more than r random bits?

Success probability amplification

It uses r random bits and takes t time.

- Consider a randomized algorithm that succeeds with probability $\frac{1}{2}$.
- Goal:** Amplify the success probability to $1 - f$.

- Set $d = \left\lceil \frac{8}{f} \right\rceil$.
- Take a d -regular Ramanujan graph with $n = 2^r$ vertices.
 - Each vertex corresponds to an r -bit string.
 - S = the set of vertices that lead to a successful execution of the algorithm.
 - $|S| = \frac{n}{2}$.
 - T = the set of vertices that do not have a neighbor in S .
 - $|S| \cdot |T| \cdot d < 4n^2 \rightarrow |T| < fn$.
- Sample a vertex v uniformly at random, for each neighbor u of v , run the algorithm using the r -bit string of u .
 - Successful $\leftrightarrow v \notin T$
 - This happens with probability at least $1 - f$.

	Time	Random bits
Standard	$O\left(t \log \frac{1}{f}\right)$	$O\left(r \log \frac{1}{f}\right)$
Ramanujan	$O\left(\frac{t}{f}\right)$	r
Next	$O\left(t \log \frac{1}{f}\right)$	$O\left(r + \log \frac{1}{f}\right)$

A Chernoff Bound for random walks

It uses r random bits and takes t time.

- Consider a randomized algorithm that succeeds with probability $\frac{1}{2}$.
- Goal:** Amplify the success probability to $1 - f$.

- Let $G = (V, E)$ be an n -vertex d -regular graph.
- Let $(v_1, v_2, \dots, v_\ell)$ be an $(\ell - 1)$ -step random walk starting from a uniformly random vertex.
- Let $f : V \rightarrow [0, 1]$ be a function.
- Let $\mu = \frac{1}{n} \sum_{v \in V} f(v)$.
- Let $\epsilon > 0$.

$$\Pr \left[\frac{1}{\ell} \sum_{i=1}^{\ell} f(v_i) \geq \mu + \epsilon + \frac{\sigma_2}{d} \right] \in e^{-\Omega(\epsilon^2 \ell)}$$

	Time	Random bits
Standard	$O\left(t \log \frac{1}{f}\right)$	$O\left(r \log \frac{1}{f}\right)$
Ramanujan	$O\left(\frac{t}{f}\right)$	r
Next	$O\left(t \log \frac{1}{f}\right)$	$O\left(r + \log \frac{1}{f}\right)$

New approach:

- Take $d \in O(1)$ and $n = 2^r$.
- Construct a random walk $(v_1, v_2, \dots, v_\ell)$ with $\ell \in O\left(\log \frac{1}{f}\right)$.
- Run the algorithm with these bit strings.

A Chernoff Bound for random walks

It uses r random bits and takes t time.

- Consider a randomized algorithm that succeeds with probability $\frac{1}{2}$.
- Goal:** Amplify the success probability to $1 - f$.

- Let $G = (V, E)$ be an n -vertex d -regular graph. $n = 2^r$
- Let $(v_1, v_2, \dots, v_\ell)$ be an $(\ell - 1)$ -step random walk starting from a uniformly random vertex. $d \in O(1)$ is large enough so that $\frac{\sigma_2}{d} \leq \frac{1}{8}$.
- Let $f : V \rightarrow [0, 1]$ be a function. 0 : the algorithm succeeds.
1 : the algorithm fails.
- Let $\mu = \frac{1}{n} \sum_{v \in V} f(v)$.
- Let $\epsilon > 0$.

$$\Pr \left[\frac{1}{\ell} \sum_{i=1}^{\ell} f(v_i) \geq \underbrace{\mu + \epsilon}_{\leq \frac{3}{4}} + \underbrace{\frac{\sigma_2}{d}}_{\leq \frac{1}{8}} \right] \in e^{-\Omega(\epsilon^2 \ell)}$$

$\frac{1}{2} \quad \frac{1}{8} \quad \leq \frac{1}{8}$

$\leq \frac{3}{4}$

$\ell \in O\left(\log \frac{1}{f}\right)$ is large enough so that the error probability $e^{-\Omega(\epsilon^2 \ell)}$ is at most f .

$\frac{1}{\ell} \sum_{i=1}^{\ell} f(v_i) < 1 \rightarrow$ some of $v_1, v_2, \dots, v_\ell$ make the algorithm successful.

Vertex connectivity

A **vertex cut** is a partition of the vertex set V into three parts:

$$V = L \cup S \cup R$$

such that $E(L, R) = \emptyset$.

The **size** of a vertex cut (L, S, R) is $|S|$.

$$L \neq \emptyset, \quad R \neq \emptyset$$

The **vertex connectivity** $\kappa(G)$ of a graph is the minimum size of a vertex cut.

$n - 1$ if G is a clique.

Vertex connectivity

A set $S \subseteq V \setminus \{s, t\}$ is an (s, t) -**vertex cut** if the removal of S disconnects s and t .

Equivalently, there is a vertex cut (L, S, R) with $s \in L$ and $t \in R$

The (s, t) -**vertex connectivity** $\kappa_G(s, t)$ is the minimum size of an (s, t) -vertex cut.

Observation: $\kappa(G) = \min_{s \neq t} \kappa_G(s, t)$.

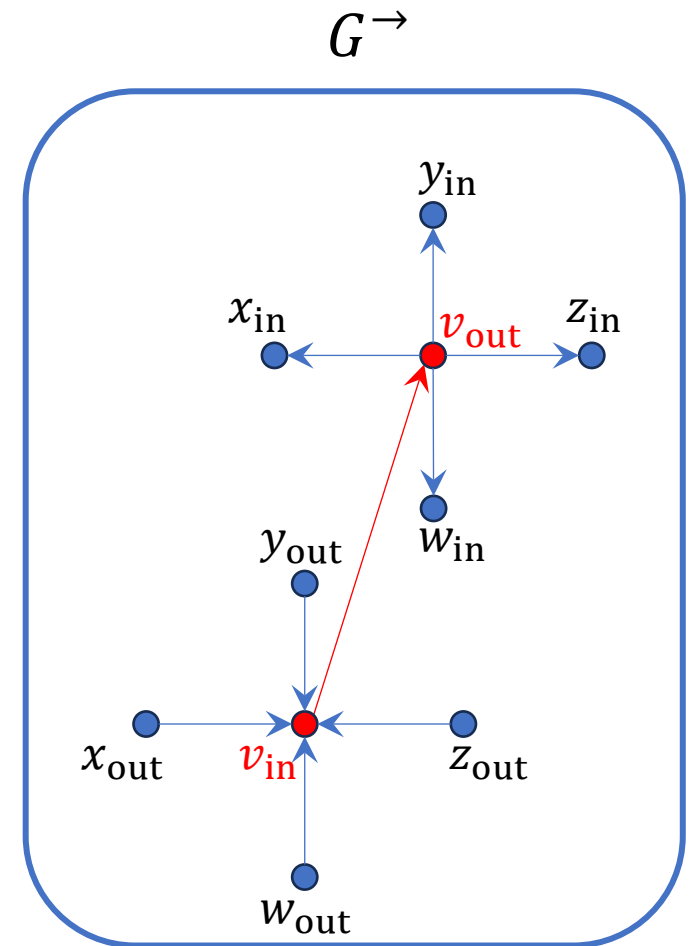
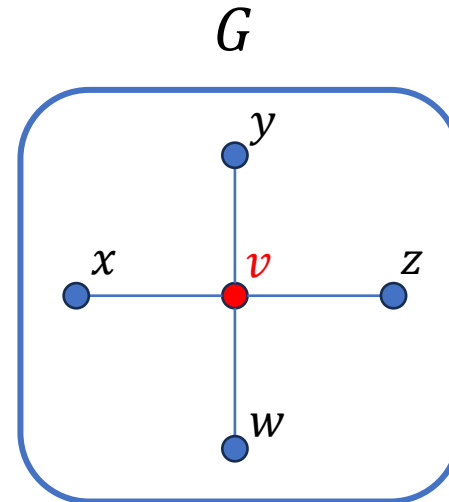
Computing a minimum (s, t) -vertex cut

- **Observation:** A minimum (s, t) -vertex cut can be computed in one **max-flow** call.

Maximum number of vertex-disjoint (s, t) -paths in G

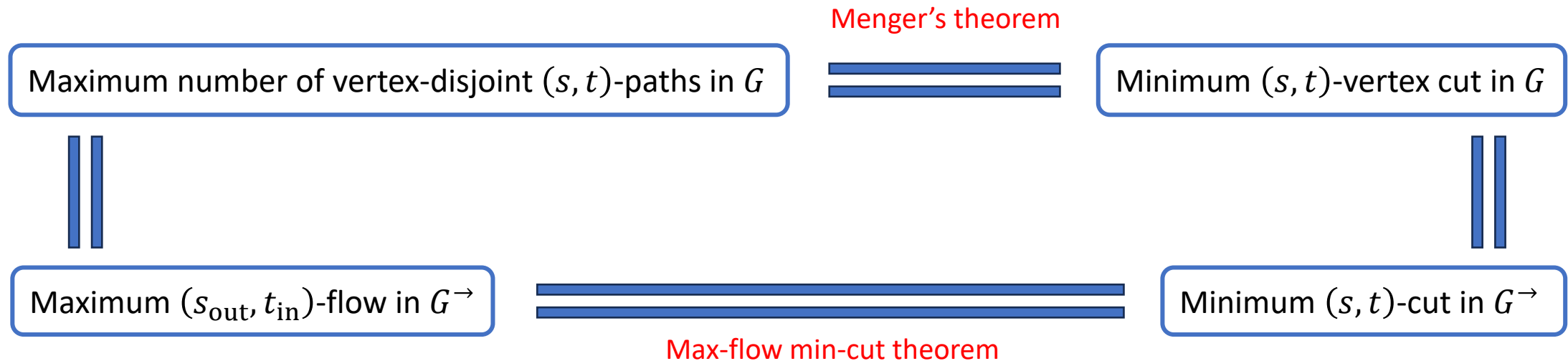


Maximum $(s_{\text{out}}, t_{\text{in}})$ -flow in $G^{\rightarrow}$



Computing a minimum (s, t) -vertex cut

- **Observation:** A minimum (s, t) -vertex cut can be computed in one **max-flow** call.



Computing a minimum vertex cut

- **Slow:** Computing a minimum (s, t) -vertex for every pair (s, t) .

This requires $O(n^2)$ max-flow calls.

- **Key observation:**

- The problem becomes easier if a minimum vertex cut is β -balanced.

A vertex cut (L, S, R) is **β -balanced** if $|L| \geq \beta$ and $|R| \in \Omega(n)$.

For $\Omega\left(\frac{\beta}{n}\right)$ fraction of the pairs (s, t) , we have: $\kappa(G) = \kappa_G(s, t)$.

By computing a minimum (s, t) -vertex cut for $O\left(\frac{n}{\beta}\right)$ random choices of (s, t) and selecting the smallest among them, we obtain a minimum vertex cut with probability at least 0.99.

Can we make it deterministic?

Derandomization via Ramanujan graphs

- We fix a β -balanced minimum vertex cut (L, S, R) of $G = (V, E)$:
 - $|L| \geq \beta$
 - $|R| \in \Omega(n)$
- **Goal:** Select $s \in L$ and $t \in R$.

Unknown to the algorithm

Consider a d -regular Ramanujan graph H over the same vertex set V .

Recall: For any $S, T \subseteq V$, if $|S| \cdot |T| \cdot d \geq 4n^2$, then $E_H(S, T) \neq \emptyset$.

There is a choice $d \in O\left(\frac{n}{\beta}\right)$ such that:

- If $|L| \geq \beta$ and $|R| \in \Omega(n)$, then $E_H(L, R) \neq \emptyset$.

This requires $nd \in O\left(\frac{n^2}{\beta}\right)$ max-flow calls.

It suffices to go over all edges $\{s, t\}$ in H .

The imbalanced case

- **Summary:** If a minimum vertex cut is β -balanced, then a minimum vertex cut can be computed using $O\left(\frac{n^2}{\beta}\right)$ max-flow calls.



Improvement over the naïve $O(n^2)$ bound.

- **Next:** Extend the approach to the imbalanced case.

The plan: gradually modify the graph to make it balanced.

The algorithm works under the condition: $\delta(G) \leq 0.99 \cdot n$

The $\delta - \kappa$ gap

- We write $\delta(G)$ to denote the minimum degree of G .

Claim: Every minimum vertex cut (L^*, S^*, R^*) of $G = (V, E)$ is β -balanced for $\beta = \delta(G) - \kappa(G)$.

By symmetry, assume $|L^*| \leq |R^*|$.

$$|R^*| \geq \frac{n - |S^*|}{2} = \frac{n - \kappa(G)}{2} \geq \frac{n - \delta(G)}{2} \geq \frac{0.01 \cdot n}{2}$$

Consider any $v \in L^*$.

$$\delta(G) \leq \deg(v) < |L^*| + |S^*| = |L^*| + \kappa(G) \rightarrow |L^*| > \delta(G) - \kappa(G) = \beta$$

The algorithm works under the condition: $\delta(G) \leq 0.99 \cdot n$

Gap amplification

- We write $\delta(G)$ to denote the minimum degree of G .

Claim: Every minimum vertex cut (L^*, S^*, R^*) of $G = (V, E)$ is β -balanced for $\beta = \delta(G) - \kappa(G)$.

The plan:

We will show that $O(n)$ max-flow calls suffice to achieve one of the following.

- A minimum vertex cut.
- Increasing the $\delta - \kappa$ gap by 1.

Total cost: $O(n^{1.5})$ max-flow calls.

$$\beta = \sqrt{n}$$

After repeating this for β times, we are in the β -balanced case.

This takes $O(\beta n)$ max-flow calls.

A minimum vertex cut can be computed using $O\left(\frac{n^2}{\beta}\right)$ max-flow calls.

Single-source vertex connectivity

- **Define:** $\kappa_G(s) = \min_{t \neq s} \kappa_G(s, t)$.

Can be computed using $n - 1$ max-flow calls.

Claim: If $\kappa_G(s) > \kappa(G)$, then $s \in S^*$ for every minimum vertex cut (L^*, S^*, R^*) of G .

If $s \notin S^*$, then $|S^*| \geq \kappa_G(s) > \kappa(G)$, so (L^*, S^*, R^*) is not a minimum cut.

Algorithm

Clarification: Since the algorithm cannot determine whether it is in Case 1 or Case 2, it records the cuts obtained in both cases. The smaller of the two is guaranteed to be a minimum cut.

- Select a vertex s .
- Compute a minimum (s, t) -vertex cut for every $t \neq s$.
- **Case 1:** $\kappa_G(s) = \kappa(G)$.
 - We have already obtained a minimum vertex cut.
- **Case 2:** $\kappa_G(s) > \kappa(G)$.
 - $s \in S^*$ for every minimum vertex cut (L^*, S^*, R^*) of G .
 - Consider $G' = G - s$.

(L^*, S^*, R^*) is a minimum cut of G



$(L^*, S^* \setminus \{s\}, R^*)$ is a minimum cut of G'

This reduces the problem of computing a minimum cut in G to the same problem in G' .

Algorithm

Recall: We want to increase the $\delta - \kappa$ gap by 1.

- Since $\kappa(G') = \kappa(G) - 1$, we achieve the goal if $\delta(G') = \delta(G)$.
- However, it is possible that $\delta(G') = \delta(G) - 1$ due to the removal of vertex s .

How to fix it?

- Select a vertex s .
- Compute a minimum (s, t) -vertex cut for every $t \neq s$.
- **Case 1:** $\kappa_G(s) = \kappa(G)$.
 - We have already obtained a minimum vertex cut.
- **Case 2:** $\kappa_G(s) > \kappa(G)$.
 - $s \in S^*$ for every minimum vertex cut (L^*, S^*, R^*) of G . $\longrightarrow \kappa(G') = \kappa(G) - 1$
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Degree restoration

- **Setting:**

- $G = (V, E)$.
- $G' = G - s$.
- $F = \{v \in V : \{v, s\} \in E, \deg_G(v) = \delta(G)\}$

How to see the existence of u ?

- **Reason 1:** The assumption that the minimum degree of the input graph is at most $0.99 \cdot n$.
- **Reason 2:** If u does not exist, then G' is already a clique.

- 
- For each $v \in F$, select a vertex $u \in V$ such that $\{u, v\} \notin E$.
 - Compute a minimum (u, v) -vertex cut in G' .
 - **Case 1:** $\kappa_{G'}(u, v) = \kappa(G')$.
 - We have already obtained a minimum vertex cut in G' .
 - **Case 2:** $\kappa_{G'}(u, v) > \kappa(G')$.

Degree restoration

- **Setting:**

- $G = (V, E)$.
- $G' = G - s$.
- $F = \{v \in V : \{v, s\} \in E, \deg_G(v) = \delta(G)\}$

$$\kappa_{G'}(u, v) > \kappa(G')$$



For every minimum cut (L^*, S^*, R^*) of G' , we have:

- $\{u, v\} \subseteq L^* \cup S^*$ or $\{u, v\} \subseteq R^* \cup S^*$.



- For each $v \in F$, select a vertex $u \in V$ such that $\{u, v\} \notin E$.
- Compute a minimum (u, v) -vertex cut in G' .
- **Case 1:** $\kappa_{G'}(u, v) = \kappa(G')$.
 - We have already obtained a minimum vertex cut in G' .
- **Case 2:** $\kappa_{G'}(u, v) > \kappa(G')$.
 - We have $\kappa(G') = \kappa(G' + \{u, v\})$.
 - Therefore, we may set: $G' \leftarrow G + \{u, v\}$.

This restore v 's degree to $\delta(G)$



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